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AT-PINN-HC: A refined time-sequential method incorporating hard-constraint strategies for predicting structural behavior under dynamic loads

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A R T I C L E I N F O

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A B S T R A C T

Physics-informed neural networks (PINNs) have been rapidly developed and offer a new computational paradigm for solving partial differential equations (PDEs) in various engineering fields. Hard constraints on boundary and initial conditions represent a significant advancement in PINNs. Given that existing hard-constraint strategies are unsuitable for structural vibration problems, this work addresses this challenge by proposing three effective hard-constraint strategies specifically for vibrational issues. Notably, the relationship between solution accuracy and the derivatives of auxiliary functions for hard constraints is identified. Based on this, various types of auxiliary functions, including polynomial, power, trigonometric, exponential, and logarithmic functions, are proposed for each hard-constraint strategy. Integrating these hard-constraint strategies and auxiliary functions into PINNs, the advanced time-marching physics-informed neural networks with hard constraints (AT-PINN-HC) are introduced. A series of numerical experiments, involving a classical Euler−Bernoulli beam, a supersonic vehicle skin panel under multi-physics loads, and a vertical standing glass plate under wind load, demonstrate that the AT-PINN-HC methods can accurately solve vibration problems in long-duration simulations. Compared to existing PINNs, AT-PINN-HC can reduce solution errors by one to four orders of magnitude and enhance training efficiency by reducing the number of iterations by up to 78 %. Additionally, the present results indicate that appropriate hard-constraint strategies and auxiliary functions must be selected on a case-by-case basis: trigonometric auxiliary functions are most effective for imposing hard constraints on boundary displacement, while exponential auxiliary functions are optimal for implementing hard constraints on initial displacement and velocity. This study not only provides effective hard-constraint strategies for vibrational problems but also provides insights into constructing hard constraints and auxiliary functions for solving other time-dependent PDEs.

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Nomenclature

AT-PINN	Advanced time-marching physics-informed neural network
AT-PINN-HC	AT-PINN with hard constraints
AT-PINN-HC1	AT-PINN with hard constraints on boundary displacement
AT-PINN-HC2	AT-PINN with hard constraints on boundary and initial displacement
AT-PINN-HC3	AT-PINN with hard constraints on boundary displacement, initial displacement and velocity
STD-PINN	Standard physics-informed neural network
a_∞	Speed of sound
c	Viscous damping coefficient
$\boldsymbol{d}_k$	Search direction of the strong Wolfe line search
D	Flexural stiffness
E	Elastic modulus
f	Frequency
g	Gravitational acceleration
g_1	Interpolation function for initial displacement in AT-PINN-HC2
g_2	Interpolation function for initial displacement in AT-PINN-HC3
g_3	Interpolation function for initial velocity in AT-PINN-HC3
h	Thickness of panel/plate
h_d, h_v	Initial displacement and velocity
$\widehat{h}_{dn}$, $\widehat{h}_{vn}$	Initial displacement and velocity of the n^{th} time segment
l	Length of beam
l_x, l_y	Dimensions of panel/plate
L	Loss function
L_{b1}, L_{b2}	Boundary displacement loss and boundary curvature loss
L_{i1}, L_{i2}	Initial displacement loss and initial velocity loss
L_p	PDE loss
L^*_{old}, L^*_{new}	Local optima of the loss function
Ma	Mach number
N_b, N_i, N_p	Number of the sample points for calculating boundary loss, initial loss and PDE loss
N_e	Number of the sample points for calculating the relative L_2 error
N_T	Thermally induced in-plane force
p	External force
$p_{acoustic}$	Acoustic load
p_{wind}	Wind load
P	Amplitude of external force
t	Time coordinate
$\widehat{t}$	Normalized time coordinate
T	Time duration
T_n	End time of the n^{th} time segment
w	Vibration displacement
w_{ref}	Reference solution
w_*	Output of STD-PINN
$\widetilde{w}_n$	Output before hard constraints in AT-PINN-HC for the n^{th} time segment
$\widehat{w}_n$	Output of AT-PINN and AT-PINN-HC for the n^{th} time segment
V	Airflow speed
x	Spatial coordinate on the x-axis
$\widehat{x}$	Normalized x coordinate
y	Spatial coordinate on the y-axis
$\widehat{y}$	Normalized y coordinate
α_{E}, α_{L}	Coefficients in ψ_{E} and ψ_{L}
α_k	Step length of the strong Wolfe line search
$\beta_{1\mathrm{E}}$, $\beta_{1\mathrm{L}}$	Coefficients in $\phi_{1\mathrm{E}}$ and $\phi_{1\mathrm{L}}$
$\beta_{2\mathrm{E}}$, $\beta_{2\mathrm{L}}$	Coefficients in $\phi_{2\mathrm{E}}$ and $\phi_{2\mathrm{L}}$
δ	Convergence coefficient of the reactivating optimization
ΔT_n	Time interval of the n^{th} time segment
ε_{L_2}	Relative L_2 error
ϕ_1	Auxiliary function for enforcing hard constraints on initial displacement
$\phi_{1\mathrm{P}}$, $\phi_{1\mathrm{T}}$, $\phi_{1\mathrm{E}}$, $\phi_{1\mathrm{L}}$	Auxiliary function ϕ_1 based on the power, trigonometric, exponential and logarithmic functions

ϕ_2	Auxiliary function for enforcing hard constraints on initial displacement and velocity
$\phi_{2\mathrm{P}}$, $\phi_{2\mathrm{T}}$, $\phi_{2\mathrm{E}}$, $\phi_{2\mathrm{L}}$	Auxiliary function ϕ_2 based on the power, trigonometric, exponential and logarithmic functions
γ_1, γ_2, γ_3	Coefficients in g_1, g_2 and g_3
λ_{b1}, λ_{b2}	Weights of boundary displacement loss and boundary curvature loss
λ_{i1}, λ_{i2}	Weights of initial displacement loss and initial velocity loss
λ_p	Weight of PDE loss
ν	Poisson's ratio
$\boldsymbol{\theta}$	Trainable parameters of the neural network
$\boldsymbol{\theta}_*$	Optimal parameters of the neural network
ρ	Density of panel/plate
ρ_a	Air density
ρ_l	Mass per unit length of beam
ξ	Amplification coefficient of the reactivating optimization algorithm
ψ	Auxiliary function for enforcing hard constraints on boundary displacement
ψ_{P}, ψ_{T}, ψ_{E}, ψ_{L}	Auxiliary function ψ based on the polynomial, trigonometric, exponential and logarithmic functions

1. Introduction

Partial differential equations (PDEs) are used to model scientific problems mathematically in various engineering fields. As a type of deep learning method, physics-informed neural networks (PINNs) have been rapidly developed in recent years, and provide a promising approach for solving PDEs. Based on the framework of deep learning, PINNs incorporate physical information, such as PDEs, boundary conditions, and initial conditions, into the neural network's loss function as regularizers. After a training process, the physical constraints can be approximately satisfied, and PINNs can learn the mapping relationship from the spatiotemporal domain to the solutions of PDEs. PINNs offer a new perspective on solving PDEs. They are meshless and have the advantages of universality and physical interpretability. To date, PINNs have been successfully applied across various scientific fields, including fluid mechanics [1–4], solid mechanics [5–7], structural mechanics [8–10], wave mechanics [11–15], biomedical science [16–18], optics [19], chemistry [20] and thermodynamics [21,22].

Raissi et al. [23] first proposed PINNs and laid the foundation of this approach. Since then, significant efforts have been devoted to enhancing the solution accuracy, training efficiency and applicability of PINNs through different approaches. These include improving sampling strategies [24,25], modifying network architectures [26–29], adopting domain decomposition techniques [30–32], and using novel activation and loss functions [33–36]. These advancements have led to various forms of PINNs, such as A-PINN for solving integro-differential equations [37], B-PINNs for PDEs with uncertainty [28], fPINNs for fractional PDEs [38], bc-PINN and AT-PINN for time-dependent PDEs [39,40].

Furthermore, the implementation of hard constraints for boundary and initial conditions represents a significant advancement in PINNs. Typically, boundary and initial conditions of PDEs are incorporated into the loss function of PINNs, it is known as "soft" constraints. This approach only approximately satisfies the conditions, as the loss function cannot be reduced to zero. To overcome this limitation, the implementation of hard constraints for boundary and initial conditions has been proposed. This technique involves multiplying the output of PINNs by an auxiliary function to satisfy homogeneous boundary and initial conditions, and then adding a simple function or neural network that meets the specified conditions. Consequently, the output with hard constraints can strictly adhere to the given boundary and initial conditions, thereby improving the solution accuracy of PINNs.

For boundary conditions, the most common and straightforward hard-constraint strategy is to construct auxiliary functions using polynomials [25,38,41–43], which are required to be zero at the boundaries. This strategy is typically used to impose hard constraints on the Dirichlet boundary condition and has been successfully applied in solving PDEs and inverse design with simple geometries. For problems with complex geometries, Wang et al. [44] constructed a radial basis function using the Gaussian function to enforce hard constraints. Sukumar and Srivastava [45] proposed a universal approximate distance function (ADF) based on R-functions, which has been validated for imposing hard constraints on the Dirichlet, Neumann and Robin boundary conditions in PINNs. Subsequently, Wang et al. [6] incorporated the ADF into their EPINN method to constrain boundary displacement, demonstrating its applicability to solid mechanics problems. In addition to closed-form auxiliary functions, hard constraints on boundary conditions can also be implemented by introducing an additional simple neural network [11,25], thereby offering better applicability but incurring higher computational costs. Research on hard constraints for initial conditions is less extensive compared to boundary conditions. Most studies have employed a simple power function to impose hard constraints on initial displacement [25,38,46,47]. Recently, Roy and Castonguay [48] presented a series of auxiliary functions based on polynomials to constrain initial conditions, achieving C^0, C^1 and C^2 temporal continuity.

This paper focuses on extending PINNs to solve vibration equations, which is crucial for addressing vibrational problems in structural dynamics. As a typical kind of time-dependent PDEs, vibration equations have fluctuant solutions in the spatiotemporal domain and often need to be solved accurately over a long duration. However, we found that the existing PINNs with hard constraints are only suitable for time-independent problems and time-dependent problems in a short duration. Additionally, the commonly used polynomial and power auxiliary functions for hard constraints do not perform well for vibrational problems, and they may even reduce the solution accuracy in some cases. To address the limitations of current PINNs and hard-constraint strategies, this study aims to introduce new auxiliary functions and hard-constraint strategies specifically for vibrational problems. These will be integrated into a series of AT-PINN-HC methods. The major novelties and contributions of this work are as follows:

- Three hard-constraint strategies for vibrational problems and the corresponding AT-PINN-HC methods are proposed, which significantly enhance solution accuracy and training efficiency compared to other existing PINNs. These strategies can also be generalized to solve other time-dependent PDEs in “long-duration” simulations.
- Multiple types of auxiliary functions for hard constraints are introduced. The relationship between solution accuracy and the derivatives of these auxiliary functions is identified, providing insights for researchers to develop novel auxiliary functions for hard constraints.
- The performance of different auxiliary functions for hard constraints in vibrational problems is evaluated. It is found that commonly used polynomial and power auxiliary functions do not perform well. The trigonometric auxiliary function is the most effective for applying hard constraints on boundary displacement, while the exponential auxiliary function is best for enforcing hard constraints on initial displacement and initial velocity.
- It is found that imposing hard constraints on additional initial conditions does not result in more accurate solutions. The highest accuracy is attained when only the boundary displacement is subjected to hard constraints.

The structure of this paper is organized as follows. The standard PINN (STD-PINN) and our proposed AT-PINN for solving vibration equations are reviewed in Sections 2.1 and 2.2, respectively. In Section 2.3, the newly proposed hard-constraint strategies, auxiliary functions and the corresponding AT-PINN-HC methods are presented. In Section 3, three illustrative examples, involving an Euler−Bernoulli beam, a skin panel of supersonic vehicles, and a standing glass plate, are analyzed. The results are compared with STD-PINN and AT-PINN to verify the effectiveness, applicability and advantages of the proposed hard constraints and AT-PINN-HC methods. Finally, the concluding remarks of this work are summarized in Section 4.

2. Research methodology

Accurately solving vibration equations is essential to analyze the dynamic behavior of structures. In this section, we use a simply supported Euler−Bernoulli beam as an example to explain the research methodologies based on PINNs and the proposed hard-constraint strategies for solving vibration equations.

2.1. Standard PINN (STD-PINN)

The PINN framework proposed by Raissi et al. [23] laid the foundation for various PINN methods, as shown in Fig. 1. In this work, it is referred to as the standard PINN method (STD-PINN).

Consider the vibration equation of a simply supported Euler−Bernoulli beam:

$$D\frac{\partial^4 w(x,t)}{\partial x^4} + c\frac{\partial w(x,t)}{\partial t} + \rho_l\frac{\partial^2 w(x,t)}{\partial t^2} = p(x,t)\,, \quad (x,t) \in (0,l) \times (0,T] \tag{1}$$

where D is the flexural stiffness, c is the viscous damping coefficient, ρ_l is the mass per unit length, $p(x,t)$ is the external force, $w(x,t)$ is the vibration displacement, l is the length of the beam, x and t are the space and time coordinates, respectively. The boundary and initial conditions of Eq. (1) are given by

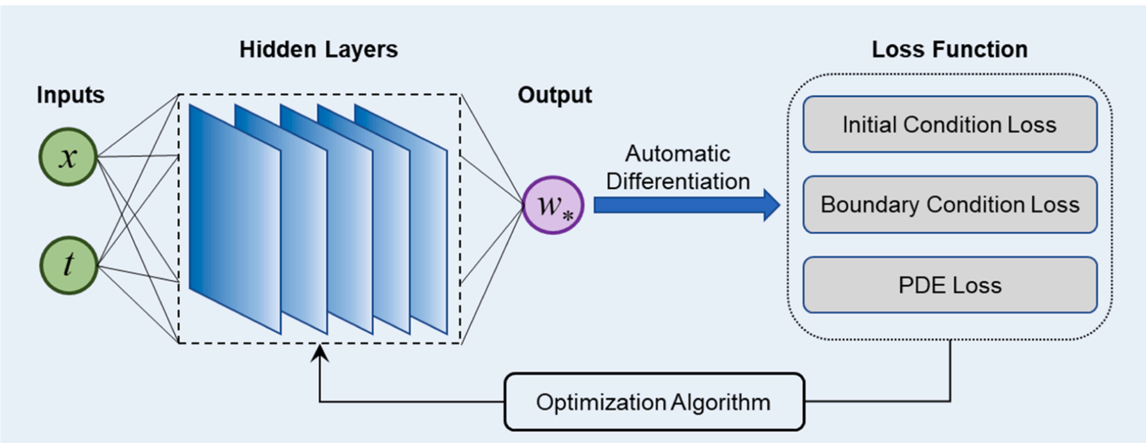

Fig. 1. Schematic diagram of the STD-PINN method.

$$\begin{cases} w(0,t) = w(l,t) = 0\,, \quad t \in [0,T] \\ \dfrac{\partial^2 w(0,t)}{\partial x^2} = \dfrac{\partial^2 w(l,t)}{\partial x^2} = 0\,, \quad t \in [0,T] \end{cases} \tag{2}$$

$$\begin{cases} w(x,0) = h_d(x)\,, \quad x \in (0,l) \\ \dfrac{\partial w(x,0)}{\partial t} = h_v(x)\,, \quad x \in (0,l) \end{cases} \tag{3}$$

where $h_d(x)$ and $h_v(x)$ are the specified initial displacement and velocity, respectively.

As shown in Fig. 1, STD-PINN solves PDEs by learning the mapping relationship between the spatiotemporal coordinates and the solution. Let $w_*(x,t,\boldsymbol{\theta})$ denote the output of the neural network, where $\boldsymbol{\theta}$ represents the trainable parameters of the neural network. w_* is a continuous differentiable function, and its derivatives with respect to x and t can be easily obtained by the automatic differentiation technique [49]. The fundamental principle of PINNs is to embed PDEs and the corresponding boundary and initial conditions into the loss function. For the vibrational problem shown in Eqs. (1)-(3), the loss function can be expressed as

$$L(\boldsymbol{\theta}) = \lambda_p L_p(\boldsymbol{\theta}) + \lambda_{b1} L_{b1}(\boldsymbol{\theta}) + \lambda_{b2} L_{b2}(\boldsymbol{\theta}) + \lambda_{i1} L_{i1}(\boldsymbol{\theta}) + \lambda_{i2} L_{i2}(\boldsymbol{\theta}) \tag{4}$$

where

$$\begin{cases} L_p(\boldsymbol{\theta}) = \dfrac{1}{N_p}\displaystyle\sum_{k=1}^{N_p}\left[\rho_l \dfrac{\partial^2 w_*(x_k^p, t_k^p)}{\partial t^2} + c\dfrac{\partial w_*(x_k^p, t_k^p)}{\partial t} + D\dfrac{\partial^4 w_*(x_k^p, t_k^p)}{\partial x^4} - p(x_k^p, t_k^p)\right]^2, \quad (x_k^p, t_k^p) \in (0,l)\times(0,T] \\ L_{b1}(\boldsymbol{\theta}) = \dfrac{1}{N_b}\displaystyle\sum_{k=1}^{N_b}\left[w_*(x_k^b, t_k^b)\right]^2, \quad (x_k^b, t_k^b) \in \{0,l\}\times[0,T] \\ L_{b2}(\boldsymbol{\theta}) = \dfrac{1}{N_b}\displaystyle\sum_{k=1}^{N_b}\left[\dfrac{\partial^2 w_*(x_k^b, t_k^b)}{\partial x^2}\right]^2, \quad (x_k^b, t_k^b) \in \{0,l\}\times[0,T] \\ L_{i1}(\boldsymbol{\theta}) = \dfrac{1}{N_i}\displaystyle\sum_{k=1}^{N_i}\left[w_*(x_k^i, t_k^i) - h_d(x_k^i)\right]^2, \quad (x_k^i, t_k^i) \in (0,l)\times\{0\} \\ L_{i2}(\boldsymbol{\theta}) = \dfrac{1}{N_i}\displaystyle\sum_{k=1}^{N_i}\left[\dfrac{\partial w_*(x_k^i, t_k^i)}{\partial t} - h_v(x_k^i)\right]^2, \quad (x_k^i, t_k^i) \in (0,l)\times\{0\} \end{cases} \tag{5}$$

Here, $L_p(\boldsymbol{\theta})$ is the PDE loss, $L_{b1}(\boldsymbol{\theta})$ is the boundary displacement loss, $L_{b2}(\boldsymbol{\theta})$ is the boundary curvature loss, $L_{i1}(\boldsymbol{\theta})$ is the initial displacement loss, and $L_{i2}(\boldsymbol{\theta})$ is the initial velocity loss. The hyperparameters λ_p, λ_{b1}, λ_{b2}, λ_{i1} and λ_{i2} are the weights of the corresponding loss components, respectively. (x_k^p, t_k^p), (x_k^b, t_k^b) and (x_k^i, t_k^i) are the spatiotemporal coordinates of the sample points for the PDE, boundary conditions and initial conditions, respectively. N_p, N_b and N_i are the corresponding number of sample points.

To minimize the loss function, the trainable parameters of the neural network can be iteratively trained through an optimization algorithm to find the optimal parameters:

$$\boldsymbol{\theta}_* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\ L(\boldsymbol{\theta}) \tag{6}$$

When the loss value is reduced to be sufficiently low, the output of the neural network $w_*(x,t)$ can approximately satisfy the PDE, the boundary and initial conditions, so that it can be regarded as the approximate solution. To estimate the accuracy of w_*, the relative L_2 error is usually used to compare it with the reference solution w_{ref}:

$$\varepsilon_{L_2} = \frac{\sqrt{\frac{1}{N_e}\sum_k^{N_e}\left[w_*(x_k^e, t_k^e) - w_{ref}(x_k^e, t_k^e)\right]^2}}{\sqrt{\frac{1}{N_e}\sum_k^{N_e}\left[w_{ref}(x_k^e, t_k^e)\right]^2}}, \quad (x_k^e, t_k^e) \in (0,l)\times[0,T] \tag{7}$$

where N_e and (x_k^e, t_k^e) are the number and coordinates of the sample points for calculating the relative L_2 error, respectively.

2.2. Advanced time-marching PINN (AT-PINN)

The STD-PINN method often suffers from large errors when solving time-dependent PDEs in “long-duration” simulations for structural analysis [40]. As the simulation duration extends, the mapping relationship between the spatiotemporal coordinates and the solution becomes more complicated, and the spatiotemporal domain for sampling progressively expands. This makes it more difficult to train the neural network and thus results in inaccurate solutions. As a type of time-dependent PDEs, vibration equations are always required to be accurately solved in a long duration. To resolve this problem, we recently proposed an advanced time-marching physics-informed neural network (AT-PINN) [40] as shown in Fig. 2. In AT-PINN, the entire temporal domain is divided into multiple sequential time segments:

$$[0, T_1],\ [T_1, T_2],\ \dots,\ [T_{n-1}, T_n],\ \dots,\ [T_{M-1}, T] \tag{8}$$

Instead of solving PDEs in the entire spatiotemporal domain as STD-PINN, the AT-PINN method solves PDEs in one time segment at a time, and the predicted solution at the end of each time segment is applied to the next time segment as the initial conditions. In this way, PDEs can be successively solved over time segments until the solution in the entire spatiotemporal domain is acquired. Compared with other PINNs training neural networks in decomposed time segments [39,47,50,51], the AT-PINN method adopts four key techniques to reduce the accumulative error and improve the training efficiency, so that it can maintain excellent accuracy in long-duration simulations. These techniques are the normalization of the spatiotemporal domain, a reactivating optimization algorithm, transfer learning and the sine activation function.

The increase in the time coordinate of sample points can induce difficulty to train the neural network and thus lead to precision degradation. To avoid this problem, the spatiotemporal domain in each time segment is normalized to a unit domain by the min-max normalization approach [52–54]:

$$\begin{cases} \hat{x} = \dfrac{x - x_{\min}}{x_{\max} - x_{\min}} = \dfrac{x}{l} \\ \hat{t} = \dfrac{t - t_{\min}}{t_{\max} - t_{\min}} = \dfrac{t - T_{n-1}}{\Delta T_n} \end{cases} \tag{9}$$

where $\Delta T_n = T_n - T_{n-1}$ is the time interval of the n^{th} time segment, $\hat{x}$ and $\hat{t}$ are the normalized space and time coordinates, respectively. Let $\hat{w}_n(\hat{x}, \hat{t})$ denotes the output of AT-PINN in the n^{th} time segment. According to $w_*(x,t) = \hat{w}_n(\hat{x}, \hat{t})$ and $\frac{\partial^{r+s}}{\partial x^r \partial t^s} = \frac{\partial^{r+s}}{\partial \hat{x}^r \partial \hat{t}^s} l^{-r} \Delta T_n^{-s}$, the loss function components of AT-PINN in the n^{th} time segment are expressed as

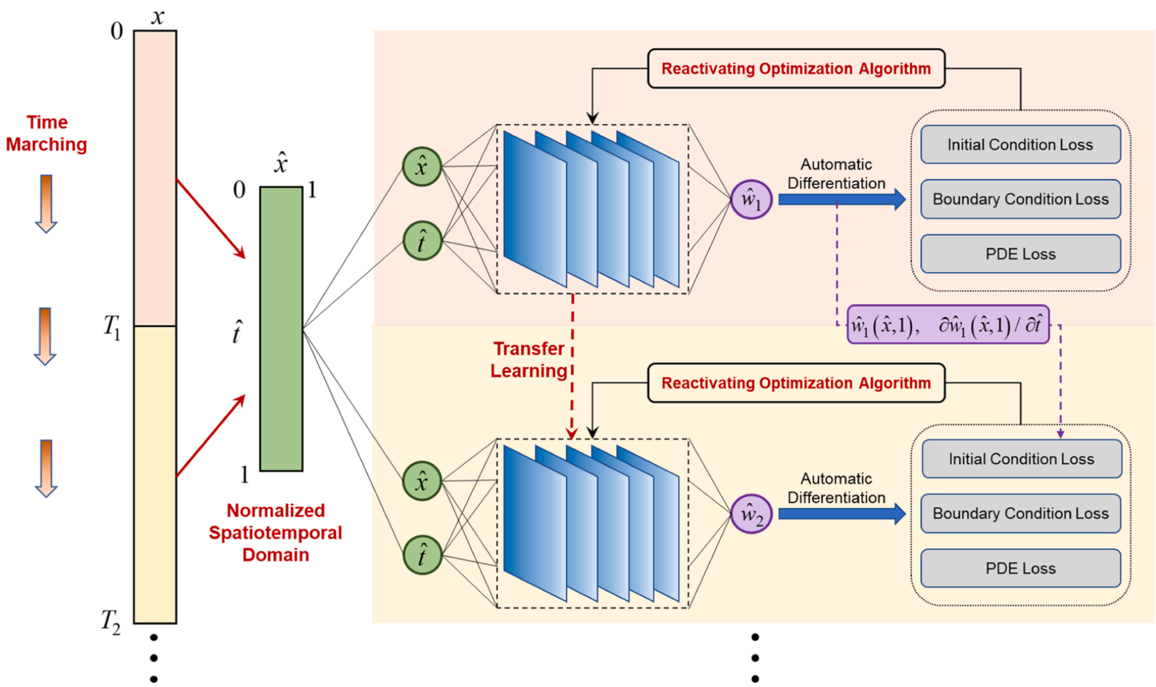

Fig. 2. Schematic diagram of the AT-PINN method.

$$
\begin{cases}
L_p(\boldsymbol{\theta}) = \dfrac{1}{N_p}\displaystyle\sum_{k=1}^{N_p}\left[\dfrac{\rho_l}{\Delta T_n^2}\dfrac{\partial^2 \widehat{w}_n(\widehat{x}_k^p,\widehat{t}_k^p)}{\partial t^2} + \dfrac{c}{\Delta T_n}\dfrac{\partial \omega_n(\widehat{x}_k^p,\widehat{t}_k^p)}{\partial \widehat{t}} + \dfrac{D}{l^4}\dfrac{\partial^4 \widehat{w}_n(\widehat{x}_k^p,\widehat{t}_k^p)}{\partial x^4} - p(\widehat{x}_k^p l, \widehat{t}_k^p \Delta T_n + T_{n-1})\right]^2, \ (\widehat{x}_k^p,\widehat{t}_k^p) \in (0,1)\times(0,1] \\
L_{b1}(\boldsymbol{\theta}) = \dfrac{1}{N_b}\displaystyle\sum_{k=1}^{N_b}\left[\widehat{w}_n(\widehat{x}_k^b,\widehat{t}_k^b)\right]^2, \ (\widehat{x}_k^b,\widehat{t}_k^b) \in \{0,1\}\times[0,1] \\
L_{b2}(\boldsymbol{\theta}) = \dfrac{1}{N_b}\displaystyle\sum_{k=1}^{N_b}\left[\dfrac{\partial^2 \widehat{w}_n(\widehat{x}_k^b,\widehat{t}_k^b)}{\partial \widehat{x}^2}\right]^2, \ (\widehat{x}_k^b,\widehat{t}_k^b) \in \{0,1\}\times[0,1] \\
L_{i1}(\boldsymbol{\theta}) = \dfrac{1}{N_i}\displaystyle\sum_{k=1}^{N_i}\left[\widehat{w}_n(\widehat{x}_k^i,\widehat{t}_k^i) - \widehat{h}_{dn}(\widehat{x}_k^i)\right]^2, \ (\widehat{x}_k^i,\widehat{t}_k^i) \in (0,1)\times\{0\} \\
L_{i2}(\boldsymbol{\theta}) = \dfrac{1}{N_i}\displaystyle\sum_{k=1}^{N_i}\left[\dfrac{\partial \widehat{w}_n(\widehat{x}_k^i,\widehat{t}_k^i)}{\partial \widehat{t}} - \widehat{h}_{vn}(\widehat{x}_k^i)\right]^2, \ (\widehat{x}_k^i,\widehat{t}_k^i) \in (0,1)\times\{0\}
\end{cases}
\tag{10}
$$

where $\widehat{h}_{dn}(\widehat{x}_k^i)$ and $\widehat{h}_{vn}(\widehat{x}_k^i)$ are the initial displacement and velocity of the n^{th} time segment. In the first time segment, they are determined by the specified initial conditions h_d and h_v given in Eq. (3):

$$
\begin{cases}
\widehat{h}_{d1}(\widehat{x}_k^i) = h_d(x_k^i) \\
\widehat{h}_{v1}(\widehat{x}_k^i) = h_v(x_k^i)\Delta T
\end{cases}
\tag{11}
$$

In the n^{th} time segment, they are determined by the results of the $(n\text{-}1)^{\text{th}}$ time segment:

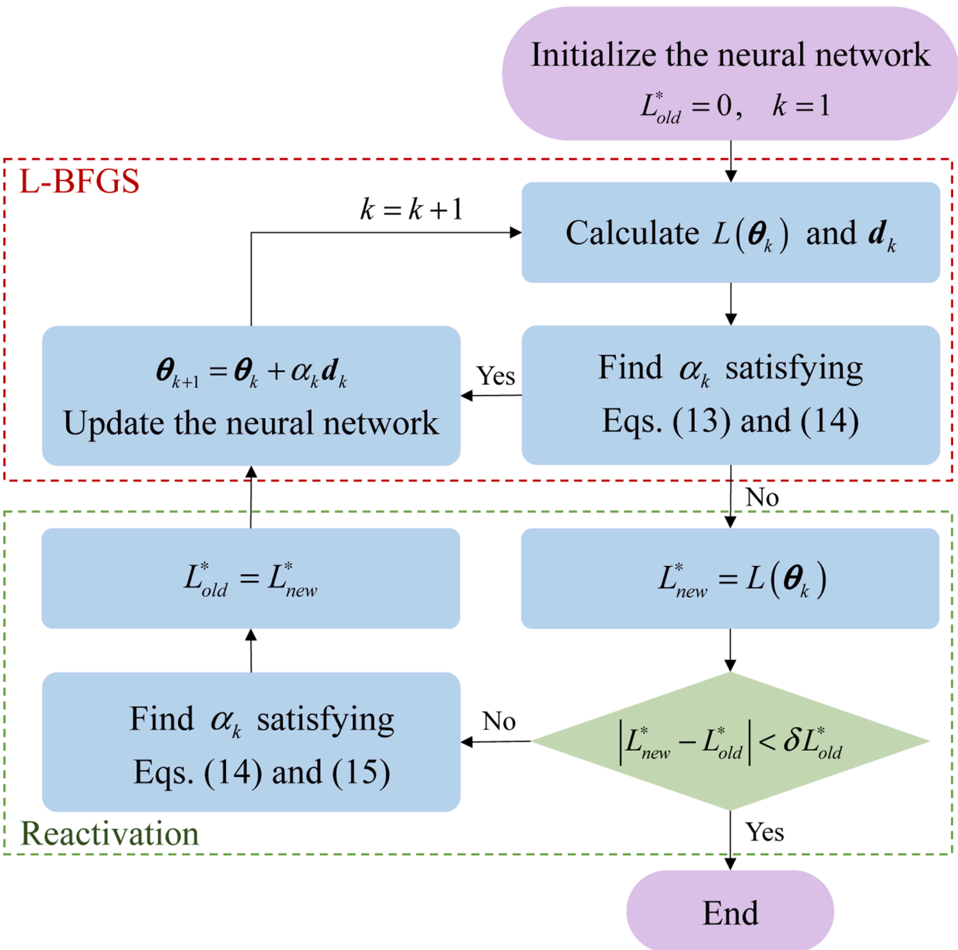

Fig. 3. Flowchart of the reactivating optimization algorithm used in AT-PINN.

$$\begin{cases} \widehat{h}_{dn}\left(\widehat{x}_k^i\right) = \widehat{w}_{n-1}\left(\widehat{x}_k^i, 1\right) \\ \widehat{h}_{vn}\left(\widehat{x}_k^i\right) = \dfrac{\partial \widehat{w}_{n-1}\left(\widehat{x}_k^i, 1\right)}{\partial \widehat{t}} \end{cases} \tag{12}$$

Using PINNs entails solving a minimization problem, as stated in Eq. (6). To avoid the presence of a locally optimal solution that is far away from the exact solution, a reactivating optimization algorithm based on the L-BFGS optimizer [55] with the strong Wolfe line search [56] is introduced into the AT-PINN method. In the k^{th} iteration of the L-BFGS optimizer, the standard strong Wolfe conditions are expressed as

$$L(\boldsymbol{\theta}_k + \alpha_k \boldsymbol{d}_k) \leq L(\boldsymbol{\theta}_k) + c_1 \alpha_k \boldsymbol{d}_k^{\mathrm{T}} \nabla L(\boldsymbol{\theta}_k) \tag{13}$$

$$\left|\boldsymbol{d}_k^{\mathrm{T}} \nabla L(\boldsymbol{\theta}_k + \alpha_k \boldsymbol{d}_k)\right| \leq c_2 \left|\boldsymbol{d}_k^{\mathrm{T}} \nabla L(\boldsymbol{\theta}_k)\right| \tag{14}$$

where α_k is the step length, $\boldsymbol{d}_k$ is the search direction, c_1 and c_2 are the constant coefficients. When there are no values of α_k that satisfy Eqs. (13) and (14), the L-BFGS optimizer is trapped in a local optimum. To reactivate the optimization process, a loose condition instead of Eq. (13) is introduced:

$$L(\boldsymbol{\theta}_k + \alpha_k \boldsymbol{d}_k) \leq (1+\xi) L(\boldsymbol{\theta}_k) + c_1 \alpha_k \boldsymbol{d}_k^{\mathrm{T}} \nabla L(\boldsymbol{\theta}_k) \tag{15}$$

where ξ is an amplification coefficient slightly larger than zero. It is easier to find a suitable α_k satisfying Eqs. (14) and (15), so that the optimization process can escape from the local optimum. Let L_{old}^* and L_{new}^* denote the loss function values associated with two successive local optima. The reactivating optimization algorithm stops when $|L_{new}^* - L_{old}^*| < \delta L_{old}^*$, where δ is a convergence coefficient much less than one. The flow chart of the reactivating optimization algorithm is shown in Fig. 3.

After the neural network is trained in the n^{th} time segment, the trained neural network is used as the initial neural network in the $(n+1)^{\text{th}}$ time segment as shown in Fig. 2, which can remarkably reduce the training time. This is inspired by the transfer learning technique and the similarity between solutions in adjacent time segments. In addition, the sine function is used as the activation function in AT-PINN, which has advantages in solution accuracy and training efficiency for solving time-dependent PDEs due to the periodicity of the sine function and its derivatives.

2.3. Advanced time-marching PINN with hard constraints (AT-PINN-HC)

In PINN methods, both boundary and initial conditions are usually embedded in the loss function. The loss function is reduced by training the neural network so that the boundary and initial conditions can be approximately satisfied, which is termed soft constraints. Since the loss function usually cannot be reduced to zero, the boundary and initial conditions cannot be strictly met by soft constraints. In some complex problems where it is difficult to train the neural network and reduce the loss function, it may result in large errors in the initial and boundary solutions. To resolve this problem, the hard constraints on boundary and initial conditions have been studied by modifying the output of the neural network with auxiliary functions [6,25,38,41–48,57]. The modified output after hard constraints can strictly meet the boundary and initial conditions, which is helpful to improve the accuracy of solving PDEs by PINNs.

In addition, the loss function of PINNs is composed of multiple components corresponding to PDEs, boundary and initial conditions. The weights of the loss function components play a crucial role in training the neural network and have a great impact on solution accuracy and training efficiency. Although some methods for adaptive adjustment of loss function weights have been studied [58–61], choosing these weights manually by trial and error usually results in more accurate solutions. Hard constraints ensure that boundary and initial conditions are automatically satisfied, eliminating the need for certain loss function components related to these conditions. Consequently, hard constraints can also reduce the complexity and effort required to adjust the loss function weights.

However, most of the existing PINNs with hard constraints are focused on time-independent problems and time-dependent problems in short duration, they are not applicable to long-duration simulations such as vibrational problems studied in this paper. Moreover, the existing hard constraints, especially some for initial conditions, are mainly based on auxiliary functions constructed by polynomial and power functions. We found that they do not perform well for solving vibration equations in terms of accuracy. Therefore, three hard-constraint strategies for vibrational problems are constructed in the present work, and different types of auxiliary functions are studied to identify the most suitable one.

In this work, three effective strategies are proposed by imposing:

(i) hard constraints on boundary displacement;
(ii) hard constraints on boundary and initial displacement; and
(iii) hard constraints on boundary displacement, initial displacement and velocity.

This research builds on the authors' recent work [40], resulting in three new advanced time-sequential PINNs with hard constraints, collectively termed AT-PINN-HC, as illustrated in Fig. 4. Corresponding to the three hard-constraint strategies, the proposed methods are designated as AT-PINN-HC1, AT-PINN-HC2, and AT-PINN-HC3, respectively. These methods can solve time-dependent problems over long durations and offer improved solution accuracy and training efficiency compared to AT-PINN.

2.3.1. AT-PINN-HC1: hard constraints on boundary displacement

The schematic diagram of the proposed AT-PINN-HC1 in the n^{th} time segment is illustrated in Fig. 4(a), where $\widetilde{w}_n$ denotes the direct output of the neural network before the implementation of hard constraints, and $\widehat{w}_n$ denotes the output after hard constraints. To impose hard constraints on the boundary displacement given in Eq. (2), $\widehat{w}_n$ are calculated by modifying $\widetilde{w}_n$ with $\psi(\widehat{x})$:

$$\widehat{w}_n(\widehat{x},\widehat{t}) = \psi(\widehat{x})\widetilde{w}_n(\widehat{x},\widehat{t}) \tag{16}$$

where $\psi(\widehat{x})$ is the auxiliary function for enforcing hard constraints on the boundary displacement, which is required to meet $\psi(0) = \psi(1) = 0$. As widely adopted in the literature [25,38,41–43], $\psi(\widehat{x})$ can be easily constructed by a polynomial:

$$\psi_{\mathrm{P}}(\widehat{x}) = 4\widehat{x}(1-\widehat{x}) = 4\widehat{x} - 4\widehat{x}^2 \tag{17}$$

In addition, we present three different types of auxiliary functions for enforcing hard constraints on the boundary displacement, which are constructed by a trigonometric function, an exponential function and a logarithmic function, respectively:

$$\psi_{\mathrm{T}}(\widehat{x}) = \sin(\pi\widehat{x}) \tag{18}$$

$$\psi_{\mathrm{E}}(\widehat{x}) = \left(1-\mathrm{e}^{\alpha_{\mathrm{E}}\hat{x}}\right)\left[1-\mathrm{e}^{\alpha_{\mathrm{E}}(1-\hat{x})}\right]\Big/\left(1-\mathrm{e}^{\alpha_{\mathrm{E}}/2}\right)^2 \tag{19}$$

$$\psi_{\mathrm{L}}(\widehat{x}) = \ln(\alpha_{\mathrm{L}}\widehat{x}+1)\ln[\alpha_{\mathrm{L}}(1-\widehat{x})+1]\Big/[\ln(\alpha_{\mathrm{L}}/2+1)]^2 \tag{20}$$

where α_{E} and α_{L} are coefficients of ψ_{E} and ψ_{L}, respectively. They are set as trainable parameters of AT-PINN-HC1 and automatically adjusted in the training process. The curves of the four auxiliary functions are presented in Fig. 5(a). Note that they are all normalized to a unit range [0,1]. By imposing hard constraints on the boundary displacement, the loss component $L_{b1}(\boldsymbol{\theta})$ in Eq. (4) vanishes, and the loss function of AT-PINN-HC1 is expressed as

$$L(\boldsymbol{\theta}) = \lambda_p L_p(\boldsymbol{\theta}) + \lambda_{b2}L_{b2}(\boldsymbol{\theta}) + \lambda_{i1}L_{i1}(\boldsymbol{\theta}) + \lambda_{i2}L_{i2}(\boldsymbol{\theta}) \tag{21}$$

2.3.2. AT-PINN-HC2: hard constraints on boundary and initial displacement

As shown in Fig. 4(b), to impose hard constraints on the boundary and initial displacement simultaneously, the output of AT-PINN-HC2 in the n^{th} time segment can be straightforward constructed as

$$\widehat{w}_n(\widehat{x},\widehat{t}) = \phi_1(\widehat{t})\psi(\widehat{x})\widetilde{w}_n(\widehat{x},\widehat{t}) + \widehat{h}_{dn}(\widehat{x}) \tag{22}$$

where $\phi_1(\widehat{t})$ is the auxiliary function for enforcing hard constraints on the initial displacement; $\widehat{h}_{dn}(\widehat{x})$ is the initial displacement of the n^{th} time segment defined in Eqs. (11) and (12). To reduce the impact of $\widehat{h}_{dn}(\widehat{x})$ on the solution far away from the initial moment, we modified Eq. (22) by introducing an interpolation function $g_1(\widehat{t})$ that decays exponentially with a trainable parameter γ_1:

$$\begin{cases} \widehat{w}_n(\widehat{x},\widehat{t}) = \phi_1(\widehat{t})\psi(\widehat{x})\widetilde{w}_n(\widehat{x},\widehat{t}) + g_1(\widehat{t})\widehat{h}_{dn}(\widehat{x}) \\ g_1(\widehat{t}) = (1-\widehat{t})\mathrm{e}^{-\gamma_1\hat{t}} \end{cases} \tag{23}$$

As shown in Fig. 5(d), $g_1(\widehat{t})$ satisfies $g_1(0) = 1$ and $g_1(1) = 0$ leading to another advantage: $\widehat{h}_{dn}(\widehat{x})$ is independent of $\widehat{h}_{d(n-1)}(\widehat{x})$ in Eq. (23), so $\widehat{w}_n(\widehat{x},\widehat{t})$ can be predicted only using the two neural networks trained in the $(n-1)^{\text{th}}$ and n^{th} time segments. In contrast, $\widehat{h}_{dn}(\widehat{x})$ is dependent on $\widehat{h}_{d(n-1)}(\widehat{x})$ when using Eq. (22). Therefore, after the neural network is trained in each time segment, to predict the solution $\widehat{w}_n(\widehat{x},\widehat{t})$ in the n^{th} time segment, all the solutions in the previous time segments must be calculated first.

The auxiliary function $\phi_1(\widehat{t})$ is required to satisfy $\phi_1(0) = 0$ and $\dot{\phi}_1(0) \neq 0$. The simplest way to construct $\phi_1(\widehat{t})$ is to use a simple power function:

$$\phi_{1\mathrm{P}}(\widehat{t}) = \widehat{t} \tag{24}$$

It has been used for solving time-dependent PDEs [25,38,46,47]. We found that the first derivative of ϕ_1 at $\widehat{t} = 0$, i.e., $\dot{\phi}_1(0)$, affects the solution accuracy of AT-PINN-HC2. To get a larger $\dot{\phi}_1(0)$ than $\dot{\phi}_{1\mathrm{P}}(0)$, three auxiliary functions for enforcing hard constraints on the initial displacement are proposed in the present work by using a trigonometric function, an exponential function and a logarithmic function with the same value range of $\phi_{1\mathrm{P}}(\widehat{t})$:

$$\phi_{1\mathrm{T}}(\widehat{t}) = \sin(\pi\widehat{t}/2) \tag{25}$$

$$\phi_{1\mathrm{E}}(\widehat{t}) = \left(1-\mathrm{e}^{-\beta_{1\mathrm{E}}\hat{t}}\right)/\left(1-\mathrm{e}^{-\beta_{1\mathrm{E}}}\right) \tag{26}$$

$$\phi_{1\mathrm{L}}(\widehat{t}) = \ln(\beta_{1\mathrm{L}}\widehat{t}+1)/\ln(\beta_{1\mathrm{L}}+1) \tag{27}$$

where $\beta_{1\mathrm{E}}$ and $\beta_{1\mathrm{L}}$ are the positive coefficients of $\phi_{1\mathrm{E}}$ and $\phi_{1\mathrm{L}}$, respectively. They are trainable parameters in AT-PINN-HC2. The curves

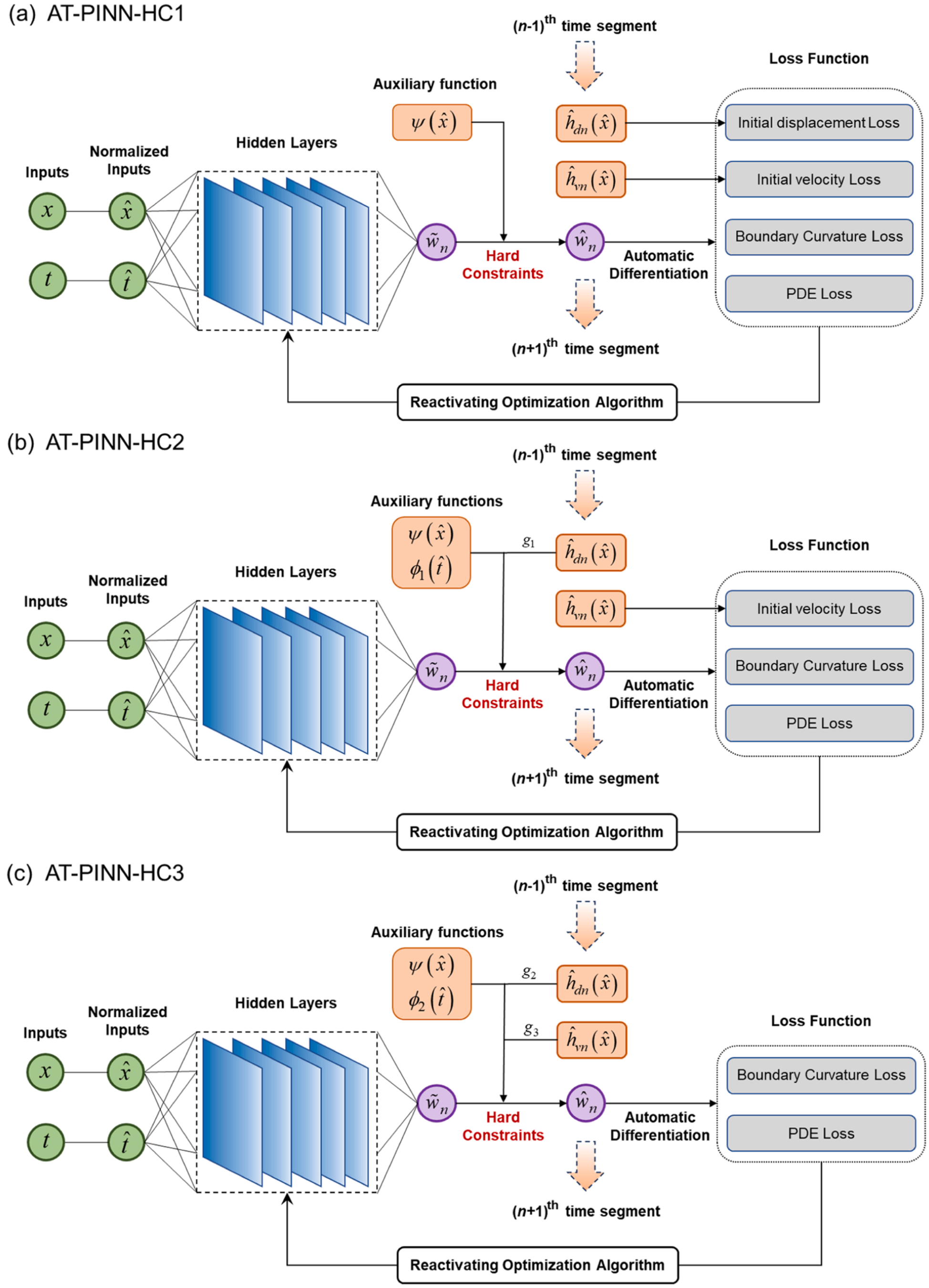

Fig. 4. Schematic diagrams of the AT-PINN-HC methods in the n^{th} time segment: (a) AT-PINN-HC1 with hard constraints on boundary displacement; (b) AT-PINN-HC2 with hard constraints on boundary and initial displacement; and (c) AT-PINN-HC3 with hard constraints on boundary displacement, initial displacement and velocity.

of the four auxiliary functions are illustrated in Fig. 5(b). Their first derivatives at $\hat{t} = 0$ are listed in Table 1. Note that the first derivatives of $\phi_{1\text{T}}$, $\phi_{1\text{E}}$ and $\phi_{1\text{L}}$ at $\hat{t} = 0$ are all larger than $\phi_{1\text{P}}$, leading to better solution accuracy of AT-PINN-HC2 than using $\phi_{1\text{P}}$. A detailed explanation will be presented in Section 3.

The boundary displacement and initial displacement are strictly satisfied through Eq. (23), so that the loss function of AT-PINN-HC2 is expressed as

$$L(\boldsymbol{\theta}) = \lambda_p L_p(\boldsymbol{\theta}) + \lambda_{b2} L_{b2}(\boldsymbol{\theta}) + \lambda_{i2} L_{i2}(\boldsymbol{\theta}) \tag{28}$$

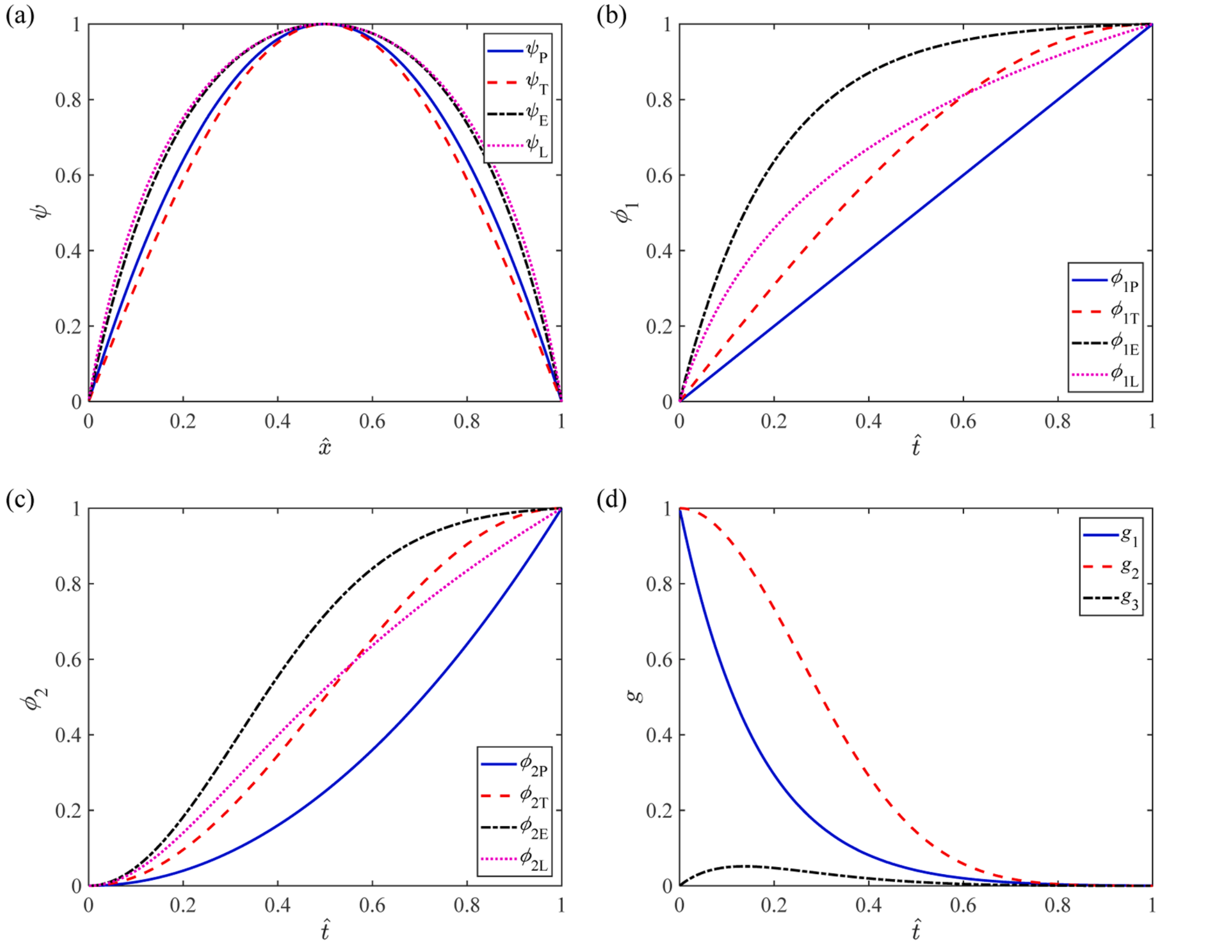

Fig. 5. (a) Auxiliary functions for hard constraints on the boundary displacement ($\alpha_E = 5$, $\alpha_L = 10$); (b) Auxiliary functions for hard constraints on the initial displacement ($\beta_{1E} = 5$, $\beta_{1L} = 10$); (c) Auxiliary functions for hard constraints on the initial displacement and velocity ($\beta_{2E} = 5$, $\beta_{2L} = 10$); (d) Interpolation functions of the initial displacement and velocity ($\gamma_1 = \gamma_2 = \gamma_3 = 5$).

2.3.3. *AT-PINN-HC3: hard constraints on boundary displacement, initial displacement and velocity*

As shown in Fig. 4(c), to enforce hard constraints on the boundary displacement, initial displacement and velocity concurrently, the output of AT-PINN-HC3 can be expressed as

$$\widehat{w}_n(\widehat{x},\widehat{t}) = \phi_2(\widehat{t})\psi(\widehat{x})\widetilde{w}_n(\widehat{x},\widehat{t}) + \widehat{h}_{dn}(\widehat{x}) + \widehat{t}\widehat{h}_{vn}(\widehat{x}) \tag{29}$$

where $\phi_2(\widehat{t})$ is the auxiliary function for hard enforcing constraints on initial displacement and velocity; $\widehat{h}_{dn}(\widehat{x})$ and $\widehat{h}_{vn}(\widehat{x})$ are the initial displacement and velocity of the n^{th} time segment defined in Eqs. (11) and (12). To make $\widehat{h}_{dn}(\widehat{x})$ and $\widehat{h}_{vn}(\widehat{x})$ independent of $\widehat{h}_{d(n-1)}(\widehat{x})$ and $\widehat{h}_{v(n-1)}(\widehat{x})$, and reduce the impact of $\widehat{h}_{dn}(\widehat{x})$ and $\widehat{h}_{vn}(\widehat{x})$ on the solution far away from $\widehat{t} = 0$, Eq. (29) is modified by introducing two interpolation functions $g_2(\widehat{t})$ and $g_3(\widehat{t})$ as follows:

$$\begin{cases} \widehat{w}_n(\widehat{x},\widehat{t}) = \phi_2(\widehat{t})\psi(\widehat{x})\widetilde{w}_n(\widehat{x},\widehat{t}) + g_2(\widehat{t})\widehat{h}_{dn}(\widehat{x}) + g_3(\widehat{t})\widehat{h}_{vn}(\widehat{x}) \\ g_2(\widehat{t}) = \left(1 - 3\widehat{t}^2 + 2\widehat{t}^3\right)e^{-\gamma_2 \hat{t}^2} \\ g_3(\widehat{t}) = \left(\widehat{t} - 2\widehat{t}^2 + \widehat{t}^3\right)e^{-\gamma_3 \hat{t}} \end{cases} \tag{30}$$

As shown in Fig. 5(d), $g_2(\widehat{t})$ and $g_3(\widehat{t})$ are selected to satisfy $g_2(0) = 1$, $g_2(1) = 0$, $\dot{g}_2(0) = \dot{g}_2(1) = 0$, $g_3(0) = g_3(1) = 0$, $\dot{g}_3(0) = 1$ and $\dot{g}_3(1) = 0$.

The auxiliary function $\phi_2(\widehat{t})$ is required to meet $\phi_2(0) = \dot{\phi}_2(0) = 0$ and $\ddot{\phi}_2(0) \neq 0$. Similar with $\phi_1(\widehat{t})$, $\phi_2(\widehat{t})$ can also be constructed by using a power function, a trigonometric function, an exponential function and a logarithmic function:

Table 1
First derivatives at $\hat{t} = 0$ of auxiliary functions for enforcing hard constraints on the initial displacement.

Auxiliary function	First derivative at $\hat{t} = 0$
ϕ_{1P}	1
ϕ_{1T}	$\pi/2$
ϕ_{1E}	$\beta_{1E}/(1 - e^{-\beta_{1E}})$
ϕ_{1L}	$\beta_{1L}/\ln(\beta_{1L} + 1)$

$$\phi_{2P}(\hat{t}) = \hat{t}^2 \tag{31}$$

$$\phi_{2T}(\hat{t}) = [1 - \cos(\pi\hat{t})]/2 \tag{32}$$

$$\phi_{2E}(\hat{t}) = \left(1 - e^{-\beta_{2E}\hat{t}^2}\right) \Big/ \left(1 - e^{-\beta_{2E}}\right) \tag{33}$$

$$\phi_{2L}(\hat{t}) = \ln\left(\beta_{2L}\hat{t}^2 + 1\right)/\ln(\beta_{2L} + 1) \tag{34}$$

where β_{2E} and β_{2L} are the positive coefficients of ϕ_{2E} and ϕ_{2L}, respectively. They are trainable parameters in AT-PINN-HC3. ϕ_{2T}, ϕ_{2E} and ϕ_{2L} are selected to have the same value range as ϕ_{2P} and larger second derivatives than ϕ_{2P} at $\hat{t} = 0$ as listed in Table 2. The curves of the four auxiliary functions are presented in Fig. 5(c).

By imposing hard constraints on the boundary displacement, initial displacement and velocity, the loss components $L_{b1}(\boldsymbol{\theta})$, $L_{i1}(\boldsymbol{\theta})$ and $L_{i2}(\boldsymbol{\theta})$ are automatically zero, and the loss function of AT-PINN-HC3 is expressed as

$$L(\boldsymbol{\theta}) = \lambda_p L_p(\boldsymbol{\theta}) + \lambda_{b2} L_{b2}(\boldsymbol{\theta}) \tag{35}$$

Compared with the loss function in Eq. (4) without hard constraints, it is much easier to adjust the weights of different loss components in Eq. (35).

3. Results and discussion

In this section, numerical experiments of the forced vibration and free vibration of a simply supported Euler−Bernoulli beam are performed by the three AT-PINN-HC methods, and the results are compared with two baseline algorithms: the STD-PINN and AT-PINN methods. Through these numerical experiments, we aim to: (i) verify the advantage of the AT-PINN-HC methods on solving vibrational problems over the baseline algorithms; (ii) compare the solution accuracy of the three AT-PINN-HC methods; and (iii) identify the best auxiliary function for the corresponding hard-constraint strategy. Moreover, the applicability of the AT-PINN-HC methods is verified by applying them to a skin panel of supersonic vehicles under multi-physics loads and a standing glass plate under wind load.

The feedforward fully connected neural network is used in all numerical experiments. The network comprises one output neuron and four hidden layers, each layer containing 32 neurons. For the vibration analysis of the beam, the network consists of two input neurons, while for the vibration analysis of the skin panel and the glass plate, it includes three input neurons. The optimization process of STD-PINN starts with the Adam optimizer for a fixed number of iterations and is followed by the L-BFGS optimizer until it converges. In the AT-PINN and AT-PINN-HC methods, the neural network is trained by the Adam optimizer and then trained by the reactivating optimization algorithm as mentioned in Section 2.2. The amplification coefficient ξ and convergence coefficient δ of the reactivating optimization algorithm are set as $\xi = 0.01$ and $\delta = 0.001$. All numerical simulations are conducted using the PyTorch 1.13.1 deep learning framework and run on an NVIDIA RTX 4090 GPU, equipped with 16,384 CUDA cores and 24 GB of memory.

3.1. Forced vibration analysis of a beam

The vibration problem of a simply supported Euler−Bernoulli beam as described in Eqs. (1)-(3) is analyzed in this section. The parameters of the beam used in this example are as follows: $l = 1$ m, $\rho_l = 1$ kg/m, $D = 2$ N · m^2, $c = 3$ N · s/m^2. The excitation force is

Table 2
Second derivatives at $\hat{t} = 0$ of auxiliary functions for enforcing hard constraints on the initial displacement and velocity.

Auxiliary function	Second derivative at $\hat{t} = 0$
ϕ_{2P}	2
ϕ_{2T}	$\pi^2/2$
ϕ_{2E}	$2\beta_{2E}/(1 - e^{-\beta_{2E}})$
ϕ_{2L}	$2\beta_{2L}/\ln(\beta_{2L} + 1)$

expressed as $p(x,t) = P\sin(\pi x/l)\cos(2\pi ft)$ with an amplitude of $P = 0.1$ N/m and a frequency of $f = 2.7$ Hz. The initial displacement $h_d(x)$ and velocity $h_v(x)$ are both set to zero. The simulation duration T is set to 20 s. For the AT-PINN and AT-PINN-HC methods, the entire time domain is divided into 20 segments, each with an equal time interval of $\Delta T = 1$ s. In each time segment, the numbers of sample points for the PDE, boundary and initial conditions are set to $N_p = 5000$, $N_b = 400$ and $N_i = 200$, respectively. For the STD-PINN method, they are set to $N_p = 100000$, $N_b = 8000$ and $N_i = 200$ by ensuring the same sampling density. All the sample points are generated by the Latin hypercube sampling. The weights of the loss components are determined through a trial-and-error process. They are given as $\lambda_p = 1$, $\lambda_{b1} = 5000$, $\lambda_{b2} = 1000$, $\lambda_{i1} = 1000$ and $\lambda_{i2} = 10$. The accuracy of each method is evaluated by the relative L_2 error, as stated in Eq. (7). It is calculated by comparing the predicted solution with the reference solution on 800,000 points uniformly sampled in the entire spatiotemporal domain, and the latter is obtained by the traditional mode superposition method and the Runge−Kutta method.

Fourteen cases with different methods and auxiliary functions are studied, and the relative L_2 errors are listed in Table 3. The total number of iterations of each case is also given in Table 3 to compare the training efficiency of different methods. In Cases 1 and 2, STD-PINN and AT-PINN are used to provide baseline results. The solutions for Cases 1 and 2 at the midpoint of the beam are illustrated in Fig. 6. The representative contours of solutions and errors for $16\text{ s} \le t \le 18$ s are given in Fig. 7. Obvious errors can be seen between the reference solution and the STD-PINN solution, which indicates the drawback of STD-PINN for long-duration vibration analysis. Our proposed AT-PINN is a powerful method for solving vibrational problems. It can reach a relative L_2 error of $O(10^{-4})$ for this example. This work targets to identify appropriate hard-constraint strategies for vibrational problems, which are anticipated to yield more accurate solutions and greater training efficiency compared to AT-PINN.

3.1.1. *Hard constraints on boundary displacement*

In Cases 3–6, AT-PINN-HC1 is used with four different auxiliary functions for enforcing hard constraints on the boundary displacement. The four results are all better than AT-PINN in terms of solution accuracy and training efficiency. Particularly, in Case 4, AT-PINN-HC1 with the trigonometric auxiliary function ψ_T achieves the most accurate solution. Its relative L_2 error is only 6.06E-05, which is smaller than STD-PINN by four orders of magnitude and smaller than AT-PINN by one order of magnitude. A satisfactory solution can also be obtained by AT-PINN-HC1 with the polynomial auxiliary function ψ_P, but it requires many more iterations than ψ_T. The representative results of Case 4 are also shown in Figs. 6 and Fig. 7, from which the excellent accuracy of AT-PINN-HC1 with ψ_T can be observed. The results of Cases 3, 5 and 6 are similar with Figs. 6(c) and Fig. 7(d) so they are not presented here for simplicity. Since the ψ_T is the best auxiliary function for the imposing of hard constraints on the boundary displacement in terms of solution accuracy and training efficiency, it is used in the following cases with AT-PINN-HC2 and AT-PINN-HC3 methods.

3.1.2. *Hard constraints on boundary and initial displacement*

In Cases 7–10, AT-PINN-HC2 is used with ψ_T and four different auxiliary functions $\phi_1(\hat{t})$ for enforcing hard constraints on the initial displacement. Although the solution accuracies of the four cases are slightly lower than Case 4 where only ψ_T is used, they are all significantly higher than STD-PINN and AT-PINN, which indicates the validity of AT-PINN-HC2. The four cases cost similar numbers of iterations, while their accuracies are different. AT-PINN-HC2 with the exponential auxiliary function ϕ_{1E} in Case 9 has the smallest relative L_2 error. To figure out the mechanism of the different performances of the four types of $\phi_1(\hat{t})$, the vibration response in the n^{th} time segment is taken for further study. According to Eq. (23), the direct output of the neural network before hard constraints can be expressed as

Table 3
Performance comparison of different methods for forced vibration analysis of a beam.

Case No.	Method	Hard constraints: Boundary displacement	Hard constraints: Initial displacement	Hard constraints: Initial velocity	Auxiliary functions	Relative L_2 error	Number of iterations
1	STD-PINN	×	×	×	N /A	1.87E-01	5.26E+04
2	AT-PINN	×	×	×	N /A	7.15E-04	1.46E+05
3	AT-PINN-HC1	√	×	×	ψ_P	9.61E-05	1.18E+05
4	AT-PINN-HC1	√	×	×	ψ_T	**6.06E-05**	**3.41E+04**
5	AT-PINN-HC1	√	×	×	ψ_E	1.70E-04	7.63E+04
6	AT-PINN-HC1	√	×	×	ψ_L	1.72E-04	6.20E+04
7	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1P}	2.23E-04	3.70E+04
8	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1T}	2.02E-04	3.68E+04
9	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1E}	**7.03E-05**	3.39E+04
10	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1L}	1.05E-04	3.46E+04
11	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2P}	2.36E-02	1.09E+05
12	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2T}	3.03E-03	7.94E+04
13	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2E}	**1.83E-04**	4.02E+04
14	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2L}	4.45E-04	4.96E+04

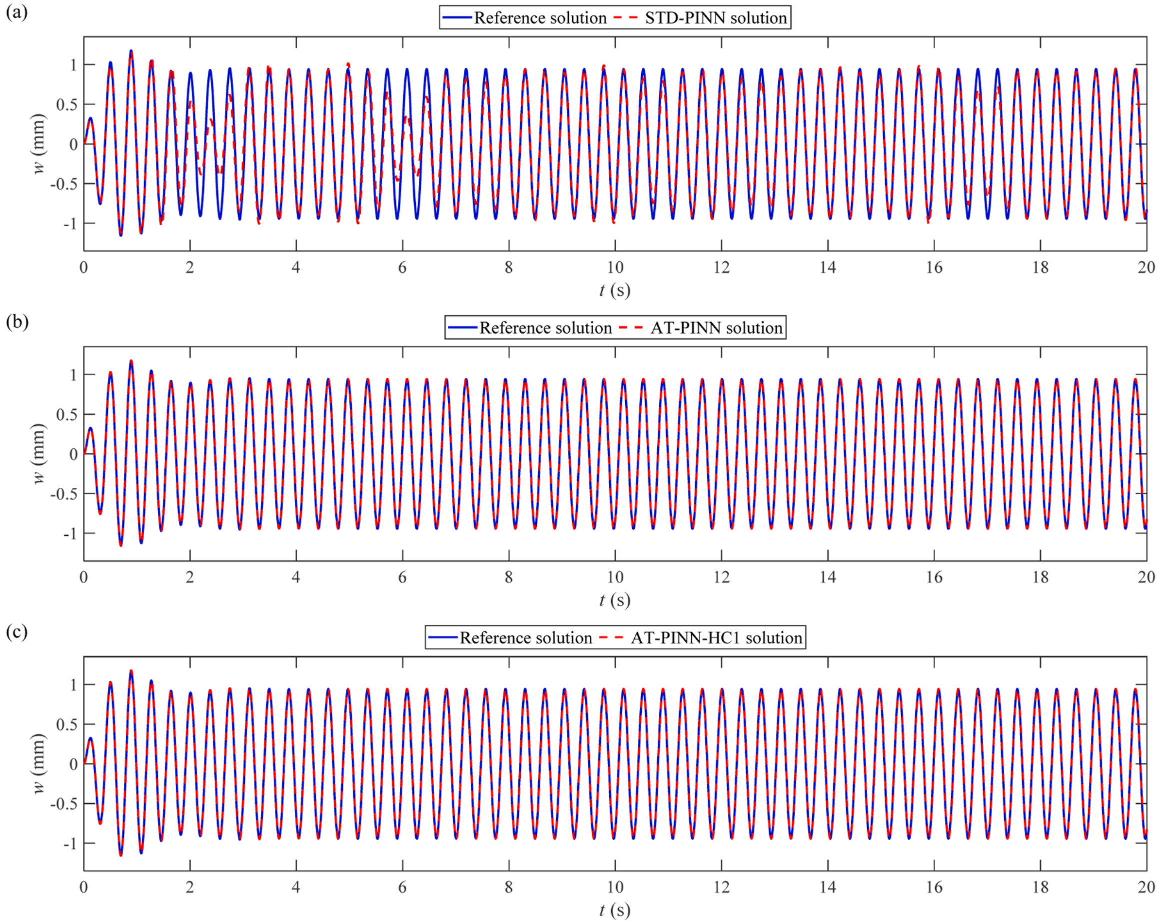

Fig. 6. Comparison of the reference solution and predicted solutions of (a) STD-PINN, (b) AT-PINN and (c) AT-PINN-HC1 with ψ_{T} at the midpoint of the forced-vibration beam.

$$\widetilde{w}_n(\widehat{x},\widehat{t}) = A(\widehat{x},\widehat{t})\frac{1}{\phi_1(\widehat{t})} \tag{36}$$

where $A(\widehat{x},\widehat{t}) = [\widehat{w}_n(\widehat{x},\widehat{t}) - g_1(\widehat{t})\widehat{h}_{dn}(\widehat{x})]/\psi(\widehat{x})$. According to Eq. (36), $1/\phi_1(\widehat{t})$ can be regarded as an amplification ratio between $\widetilde{w}_n(\widehat{x},\widehat{t})$ and $A(\widehat{x},\widehat{t})$. Because $\phi_1(\widehat{t})$ is monotonically increasing from 0 to 1 with $\widehat{t}$, the maximum amplification ratio occurs at $\widehat{t} = 0$. Therefore, the value of $\widetilde{w}_n(\widehat{x},0)$ can be regarded as an indicator to evaluate the amplification performance of different types of $\phi_1(\widehat{t})$. It is easy to find that $A(\widehat{x},\widehat{t})$ and $\phi_1(\widehat{t})$ are all infinitesimals as $\widehat{t}$ tends to 0, so $\widetilde{w}_n(\widehat{x},0)$ can be calculated by L'Hospital's rule:

$$\widetilde{w}_n(\widehat{x},0) = \lim_{\widehat{t}\to 0}\frac{A(\widehat{x},\widehat{t})}{\phi_1(\widehat{t})} = \frac{\dot{A}(\widehat{x},0)}{\dot{\phi}_1(0)} \tag{37}$$

For a given $A(\widehat{x},\widehat{t})$, the first derivative of $\phi_1(\widehat{t})$ at $\widehat{t} = 0$ determines the value of $\widetilde{w}_n(\widehat{x},0)$. Considering a simple example where $\widehat{w}_n(\widehat{x},\widehat{t}) = \psi(\widehat{x})\sin(5\pi\widehat{t})$ and $\widehat{h}_{dn}(\widehat{x}) = 0$, the values of $\widehat{w}_n(\widehat{x},\widehat{t})$ and $\widetilde{w}_n(\widehat{x},\widehat{t})$ at $\widehat{x} = 0.5$ where $\psi(\widehat{x}) = 1$ are illustrated in Fig. 8(a) and Fig. 8(b), respectively. The coefficients of $\phi_{1\mathrm{E}}$ and $\phi_{1\mathrm{L}}$ used in Fig. 8(b) are $\beta_{1\mathrm{E}} = 5$ and $\beta_{1\mathrm{L}} = 10$. Comparing the first derivatives of four different types of $\phi_1(\widehat{t})$ at $\widehat{t} = 0$ as listed in Table 1, we found that $\dot{\phi}_{1\mathrm{E}}(0) > \dot{\phi}_{1\mathrm{L}}(0) > \dot{\phi}_{1\mathrm{T}}(0) > \dot{\phi}_{1\mathrm{P}}(0)$. As a result, the absolute values of $\widetilde{w}_n(\widehat{x},0)$ with the four auxiliary functions have the corresponding magnitude relation. Moreover, $\widetilde{w}_n(\widehat{x},0)$ marks the amplification effects of $\phi_1(\widehat{t})$. As can be seen in Fig. 8(b), the larger the value of $\widetilde{w}_n(\widehat{x},0)$ is, the more dramatically $\widetilde{w}_n(\widehat{x},\widehat{t})$ changes. The neural network learns the mapping relationship between $(\widehat{x},\widehat{t})$ and $\widetilde{w}_n$. A gently varying $\widetilde{w}_n(\widehat{x},\widehat{t})$ is easy to learn, while a rapidly changing $\widetilde{w}_n(\widehat{x},\widehat{t})$ makes it difficult to train the neural network, resulting in a larger error in the solution. Therefore, the proposed exponential auxiliary function $\phi_{1\mathrm{E}}$ can achieve more accurate solutions than the power auxiliary function $\phi_{1\mathrm{P}}$.

It should be noted that the value of $\beta_{1\mathrm{E}}$ affects the performance of $\phi_{1\mathrm{E}}$. Although a larger $\beta_{1\mathrm{E}}$ results in a larger $\dot{\phi}_{1\mathrm{E}}$, an overly large $\beta_{1\mathrm{E}}$ can also cause numerical instability, so we make $\beta_{1\mathrm{E}}$ as a trainable parameter so that it can be automatically adjusted in the training process. Similarly, $\beta_{1\mathrm{L}}$ is also a trainable parameter. For Cases 7–10 listed in Table 3, the final values of $\beta_{1\mathrm{E}}$ and $\beta_{1\mathrm{L}}$ in each time segment

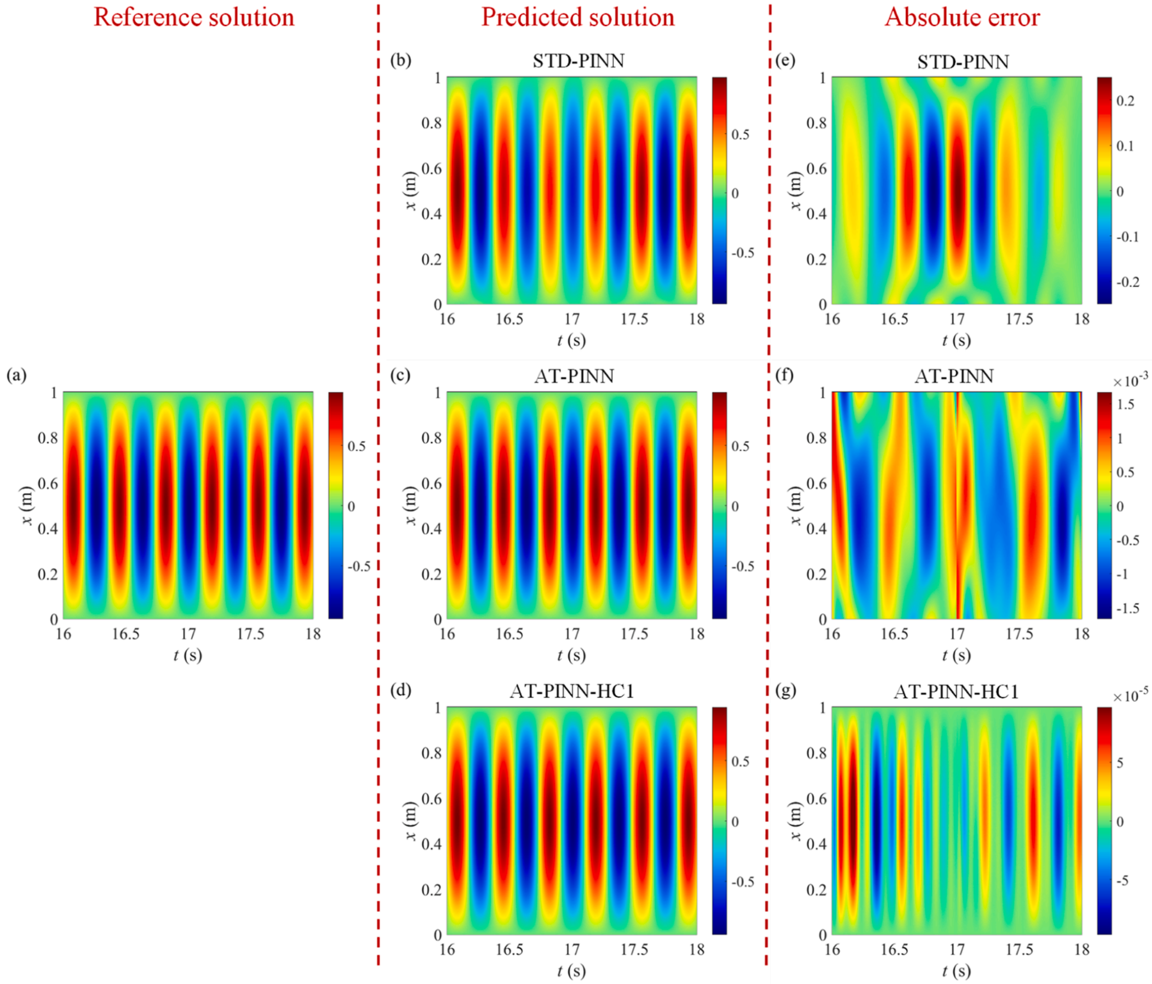

Fig. 7. Contours of solutions and errors of the forced-vibration beam when 16 s $\leq t \leq$ 18 s: (a) reference solution; (b)-(d) predicted solutions of STD-PINN, AT-PINN and AT-PINN-HC1 with ψ_{T}; (e)-(g) absolute errors of STD-PINN, AT-PINN and AT-PINN-HC1 with ψ_{T}.

are shown in Fig. 9(a), and the corresponding first derivatives of $\phi_1(\hat{t})$ at $\hat{t}=0$ are shown in Fig. 9(b). Fig. 10 compares $\widetilde{w}_n(\hat{x},\hat{t})$ with $\phi_{1\mathrm{P}}$, $\phi_{1\mathrm{T}}$, $\phi_{1\mathrm{E}}$, and $\phi_{1\mathrm{L}}$ in three representative time segments. It can be observed that $\dot{\phi}_{1\mathrm{E}}(0) > \dot{\phi}_{1\mathrm{L}}(0) > \dot{\phi}_{1\mathrm{T}}(0) > \dot{\phi}_{1\mathrm{P}}(0)$ in each time segment, so $\phi_{1\mathrm{E}}$ results in the smallest absolute value of $\widetilde{w}_n(\hat{x},0)$ and the gentlest $\widetilde{w}_n(\hat{x},\hat{t})$ as illustrated in Fig. 10. Therefore, Case 9 with $\phi_{1\mathrm{E}}$ has the most accurate results among the four cases using AT-PINN-HC2 as listed in Table 3. It indicates that the proposed exponential auxiliary function $\phi_{1\mathrm{E}}$ is the best one for enforcing hard constraints on the initial displacement.

3.1.3. Hard constraints on boundary displacement, initial displacement and velocity

In Cases 11–14, AT-PINN-HC3 is used with ψ_{T} and four different auxiliary functions $\phi_2(\hat{t})$ for hard enforcing constraints on both initial displacement and velocity. It can be found that the relative L_2 errors of the four cases are all larger than the corresponding Cases 7–10 using AT-PINN-HC2. In addition, although Cases 11–14 have lower relative L_2 errors than STD-PINN, the relative L_2 errors of Cases 11 and 12 are obviously larger than AT-PINN, which indicates the power auxiliary function $\phi_{2\mathrm{P}}(\hat{t})$ and the trigonometric auxiliary function $\phi_{2\mathrm{T}}(\hat{t})$ are not suitable for enforcing hard constraints on the initial displacement and velocity simultaneously. In contrast, the exponential auxiliary function $\phi_{2\mathrm{E}}(\hat{t})$ and logarithmic auxiliary function $\phi_{2\mathrm{L}}(\hat{t})$ perform well. The relative L_2 errors of Cases 13 and 14 are both smaller than AT-PINN, and $\phi_{2\mathrm{E}}(\hat{t})$ results in the most accurate solution among the four cases using AT-PINN-HC3. The mechanism of the different performances of the four types of $\phi_2(\hat{t})$ is similar with that discussed in Section 3.3.2. According to Eq. (30), the direct output of the neural network before hard constraints in the n^{th} time segment can be expressed as

$$\widetilde{w}_n(\hat{x},\hat{t}) = B(\hat{x},\hat{t})\frac{1}{\phi_2(\hat{t})} \tag{38}$$

where $B(\hat{x},\hat{t}) = \left[\hat{w}_n(\hat{x},\hat{t}) - g_2(\hat{t})\hat{h}_{dn}(\hat{x}) - g_3(\hat{t})\hat{h}_{vn}(\hat{x})\right]/\psi(\hat{x})$. Since $\phi_2(\hat{t})$ is monotonically increasing from 0 to 1 with $\hat{t}$ just like $\phi_1(\hat{t})$, we can still regard the value of $\tilde{w}_n(\hat{x},0)$ as the indicator to evaluate the performances of different types of $\phi_2(\hat{t})$. Because $B(\hat{x},\hat{t})$, $\phi_1(\hat{t})$ and there first derivatives are all infinitesimals as $\hat{t}$ tends to 0, so $\tilde{w}_n(\hat{x},0)$ should be calculated by using L'Hospital's rule twice:

$$\tilde{w}_n(\hat{x},0) = \lim_{\hat{t}\to 0}\frac{B(\hat{x},\hat{t})}{\phi_2(\hat{t})} = \lim_{\hat{t}\to 0}\frac{\dot{B}(\hat{x},\hat{t})}{\dot{\phi}_2(\hat{t})} = \frac{\ddot{B}(\hat{x},0)}{\ddot{\phi}_2(0)} \tag{39}$$

It means that, for a given $B(\hat{x},\hat{t})$, the second derivative of $\phi_2(\hat{t})$ at $\hat{t} = 0$ determines the value of $\tilde{w}_n(\hat{x},0)$. Fig. 11(a) shows the final values of $\beta_{2\mathrm{E}}$ and $\beta_{2\mathrm{L}}$ in each time segment after training process. Substituting them into the expressions listed in

Table 2, we can obtain the second derivatives of $\phi_{2\mathrm{P}}(\hat{t})$, $\phi_{2\mathrm{T}}(\hat{t})$, $\phi_{2\mathrm{E}}(\hat{t})$ and $\phi_{2\mathrm{L}}(\hat{t})$ at $\hat{t} = 0$ in each time segment as illustrated in Fig. 11(b). It can be found that $\ddot{\phi}_{2\mathrm{E}}(0) > \ddot{\phi}_{2\mathrm{L}}(0) > \ddot{\phi}_{2\mathrm{T}}(0) > \ddot{\phi}_{2\mathrm{P}}(0)$ in each time segment, so $\tilde{w}_n(\hat{x},0)$ with $\phi_{2\mathrm{E}}(\hat{t})$ has the smallest absolute value and the corresponding $\tilde{w}_n(\hat{x},\hat{t})$ has the gentlest varying trend as shown in Fig. 12. It means the mapping relationship between $(\hat{x},\hat{t})$ and $\tilde{w}_n$ with $\phi_{2\mathrm{E}}(\hat{t})$ is significantly easier to learn than with other types of $\phi_2(\hat{t})$, so the solution of Case 13 with $\phi_{2\mathrm{E}}(\hat{t})$ has the smallest relative L_2 error. Therefore, the proposed $\phi_{2\mathrm{E}}$ is the most suitable one among the four auxiliary functions for enforcing hard constraints on the initial displacement and velocity simultaneously.

It should be noted that the circular frequency of the vibration response in the normalized time domain is usually larger than one, so $\ddot{B}(\hat{x},0)$ can be much larger than $\dot{A}(\hat{x},0)$. It leads to that the absolute values of $\tilde{w}_n(\hat{x},0)$ in AT-PINN-HC3 are larger than those in AT-PINN-HC2, and $\tilde{w}_n(\hat{x},\hat{t})$ in AT-PINN-HC3 changes more dramatically, which can be seen by comparing the value ranges of $\tilde{w}_n(\hat{x},\hat{t})$ illustrated in Figs. 10 and Fig. 12. As a result, the mapping relationships between $(\hat{x},\hat{t})$ and $\tilde{w}_n$ in AT-PINN-HC3 are more difficult to learn so that the solution accuracies of AT-PINN-HC3 are lower than AT-PINN-HC2. Because of the similar reason, the solution accuracies of AT-PINN-HC2 are lower than AT-PINN-HC1 with ψ_{T}. It indicates that imposing hard constraints on more initial conditions may increase the complexity of the mapping relationship, making the training process of the neural network more complicated, and thus reduce the solution accuracy. However, it does not simply mean that AT-PINN-HC3 is worse than AT-PINN-HC2, or AT-PINN-HC2 is worse than AT-PINN-HC1. Using hard constraints ensures that the boundary and initial conditions are strictly satisfied, allowing for the imposition of additional constraints to reduce more components of the loss function. This, in turn, simplifies the process of adjusting the weights of the loss components. In practical applications, a trade-off of the advantages and disadvantages of hard constraints should be considered to choose the appropriate hard-constraint strategy.

The optimization process of the current AT-PINN-HC methods employs a combination of the Adam optimizer and the L-BFGS optimizer along with a reactivating optimization algorithm. To assess the effectiveness of this optimization strategy, we use Case 4 from Table 3 as an example and compare its performance with two new cases, designated as Case 15 and Case 16, which utilize only the Adam optimizer and the L-BFGS optimizer, respectively. In Cases 4 and 16, the optimization process concludes when the convergence criterion is satisfied. For Case 15, the number of iterations is set to match that of Case 4. As presented in Table 4, the maximum allocated memory and training time for Case 15 with the Adam optimizer are both slightly lower than those for Case 4, but the solution accuracy is significantly poorer. Case 16 requires the same amount of memory as Case 4 while taking less training time. However, after training with the Adam optimizer, the L-BFGS optimizer in Case 4 benefits from better initial parameters compared to Case 16, resulting in Case 4, which combines the Adam and L-BFGS optimizers, achieving the highest solution accuracy.

3.2. Free vibration analysis of a beam

The forced-vibration problem of a beam under external excitation force is studied in the above section. In this section, the excitation force is removed to verify the proposed AT-PINN-HC methods for free-vibration analysis. The initial displacement of the beam is set to $h_d(x) = \sin(\pi x/l)$ mm, and the initial velocity is set to zero. The damping coefficient of the beam is changed to $c = 0.3\ \mathrm{N\cdot s/m^2}$, while other parameters, boundary conditions, interval of time segments and number of sample points are the same as those in Section 3.1.

Similar with Section 3.1, fourteen cases with different methods and auxiliary functions are studied, and the relative L_2 errors and

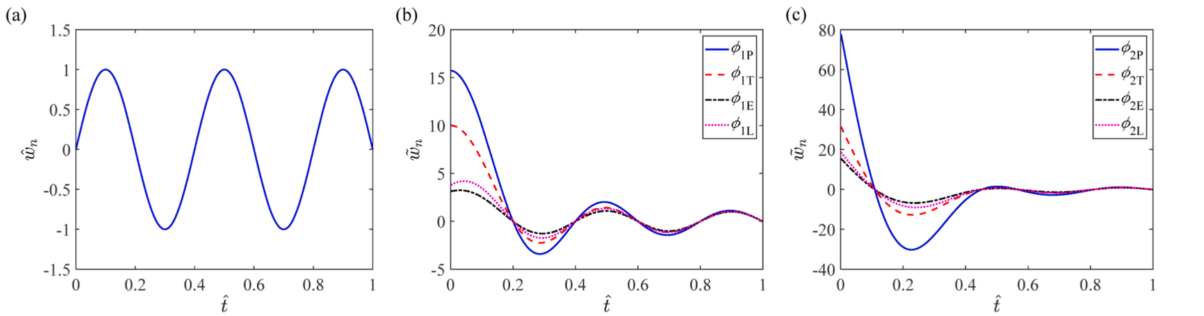

Fig. 8. Output of the neural network at $\hat{x} = 0.5$: (a) $\hat{w}_n$, output after hard constraints; (b) $\tilde{w}_n$, output before hard constraints with different types of ϕ_1 ($\beta_{1\mathrm{E}} = 5$, $\beta_{1\mathrm{L}} = 10$); (c) $\tilde{w}_n$, output before hard constraints with different types of ϕ_2 ($\beta_{2\mathrm{E}} = 5$, $\beta_{2\mathrm{L}} = 10$).

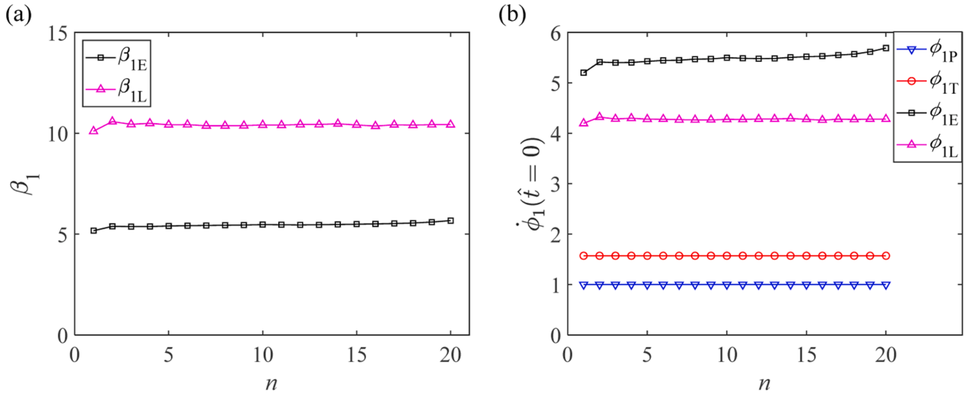

Fig. 9. (a) Final values of β_{1E} and β_{1L} in each time segment; (b) First derivatives of $\phi_{1P}(\hat{t})$, $\phi_{1T}(\hat{t})$, $\phi_{1E}(\hat{t})$ and $\phi_{1L}(\hat{t})$ at $\hat{t}=0$ in each time segment.

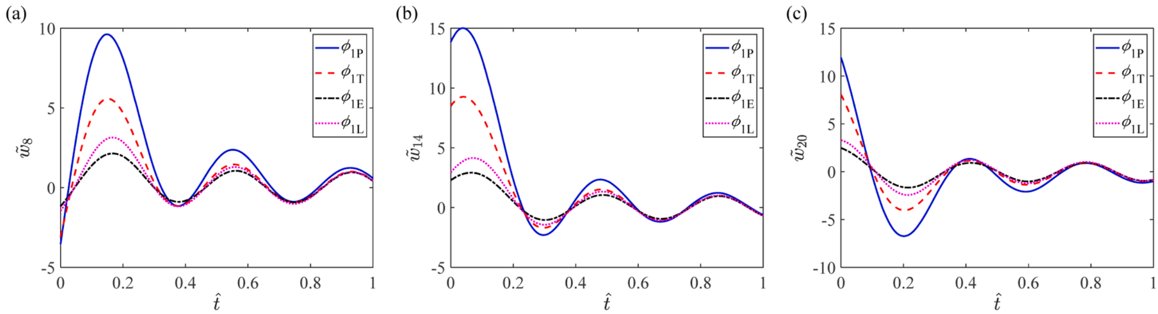

Fig. 10. Comparison of the output $\tilde{w}_n$ before hard constraints with different types of ϕ_1 in the (a) 8th, (b) 14th and (c) 20th time segments at the midpoint of the forced-vibration beam by using AT-PINN-HC2.

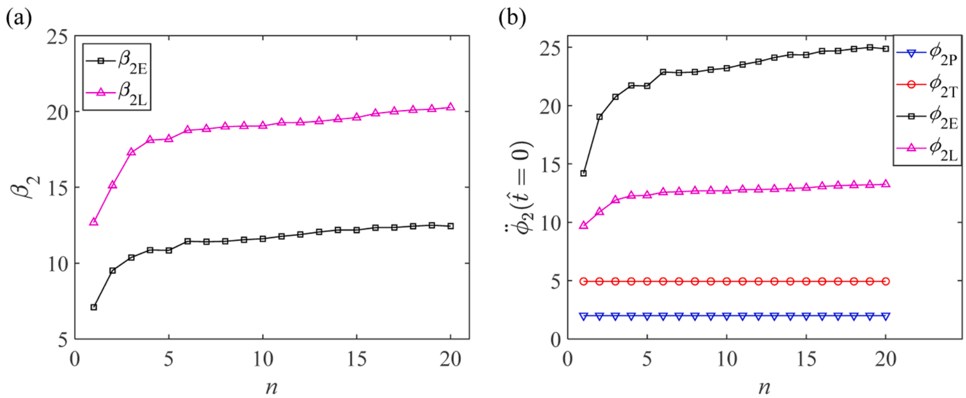

Fig. 11. (a) Final values of β_{2E} and β_{2L} in each time segment; (b) Second derivatives of $\phi_{2P}(\hat{t})$, $\phi_{2T}(\hat{t})$, $\phi_{2E}(\hat{t})$ and $\phi_{2L}(\hat{t})$ at $\hat{t}=0$ in each time segment.

numbers of iterations are compared in Table 5. Among these cases, Case 4 using AT-PINN-HC1 with ψ_T has the most accurate solution with a relative L_2 error of 2.21E-04. Fig. 13 shows the solutions at the midpoint of the beam in Case 4 and the two baseline Cases 1 and 2. The representative contours of solutions and errors for 5 s $\leq t \leq$ 7 s are given in Fig. 14. As shown in Fig. 13(a), the STD-PINN solution decreases rapidly with time and converges to zero after t = 3 s. It indicates that STD-PINN fails to predict acceptable solutions for long-duration free-vibration analysis. In contrast, the AT-PINN solution agrees well with the reference solution, and reaches a relative L_2 error of 3.65E-03. Among the four cases using AT-PINN-HC1, Case 4 with ψ_T has the smallest relative L_2 error and number of iterations, which are both smaller than those of AT-PINN by one order of magnitude. Therefore, ψ_T is used to impose hard constraints on the boundary displacement in Cases 7–14 with AT-PINN-HC2 and AT-PINN-HC3 methods. Among the four cases using AT-PINN-HC2, Case 9 with ϕ_{1E} has the smallest relative L_2 error. Among the four cases using AT-PINN-HC3, Case 13 with ϕ_{2E} has the smallest

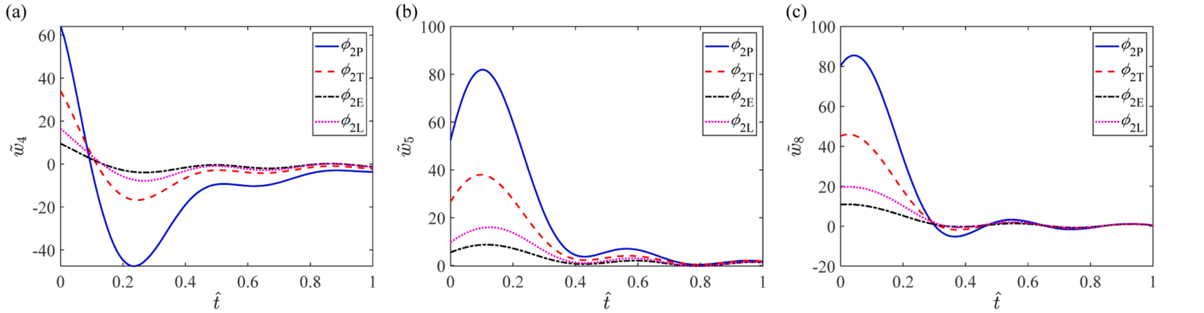

Fig. 12. Comparison of the output $\tilde{w}_n$ before hard constraints with different types of ϕ_2 in the (a) 4th, (b) 5th and (c) 8th time segments at the midpoint of the forced-vibration beam by using AT-PINN-HC3.

relative L_2 error, and its solution accuracy is higher than AT-PINN, whereas the solution accuracies with other three types of ϕ_2 are all lower than AT-PINN. The reason for the good performances of ϕ_{1E} and ϕ_{2E} has been explained in Section 3.1: the first derivative of ϕ_{1E} at $\hat{t} = 0$ is larger than other types of ϕ_1, and the second derivative of ϕ_{2E} at $\hat{t} = 0$ is larger than other types of ϕ_2. As a result, ϕ_{1E} and ϕ_{2E} can lead to gently varying $\tilde{w}_n$ as shown in Figs. 15 and Fig. 16, which can reduce the difficulty of training the neural network and thus reduce the errors of solutions. These results indicate that it is best to choose ψ_T to impose hard constraints on the boundary displacement in AT-PINN-HC1, select ϕ_{1E} to apply hard constraints on the initial displacement in AT-PINN-HC2, and opt for ϕ_{2E} to set hard constraints on both the initial displacement and velocity in AT-PINN-HC3.

3.3. Vibration analysis of a supersonic vehicle skin panel

In Sections 3.1 and 3.2, three hard-constraint strategies are studied and the most suitable auxiliary function for each strategy is found. In this section, AT-PINN-HC1 (with ψ_T), AT-PINN-HC2 (with ψ_T and ϕ_{1E}), and AT-PINN-HC3 (with ψ_T and ϕ_{2E}) are applied to solve the vibration problem of a supersonic vehicle skin panel to verify the applicability of the three AT-PINN-HC methods.

As shown in Fig. 17, the skin panel of supersonic vehicles is a typical model widely studied in structural dynamics and aerospace fields. It is usually subjected to aerodynamic, thermal and acoustic loads, and suffers from complex vibration response. Violent vibration can cause fatigue damage to the skin panel, which in turn impacts the flight performance, structural integrity, and reliability of supersonic vehicles [62–65]. The motion equation of the panel with a small deformation can be expressed as

$$\begin{cases} \dfrac{Eh^3}{12(1-\nu^2)}\nabla^4 w + N_T\nabla^2 w + \dfrac{\rho_a V^2}{\sqrt{Ma^2-1}}\left[\dfrac{\partial w}{\partial x} + \dfrac{Ma^2-2}{(Ma^2-1)V}\dfrac{\partial w}{\partial t}\right] + c\dfrac{\partial w}{\partial t} + \rho h\dfrac{\partial^2 w}{\partial t^2} = p_{acoustic} \\ (x,y,t) \in (0,l_x) \times (0,l_y) \times (0,T] \end{cases} \tag{40}$$

where E is the elastic modulus, h is the panel thickness, ν is the Poisson's ratio, N_T is the thermally induced in-plane force, ρ_a is the air density, V is the airflow speed, $Ma = V/a_\infty$ is the Mach number, a_∞ is the speed of sound, c is the viscous damping coefficient, ρ is the panel density, $p_{acoustic}$ is the acoustic load, l_x and l_y are the side lengths of the panel. In this example, these parameters are given as E = 110 GPa, h = 3 mm, ν = 0.34, c = 200 $\mathrm{N\cdot s/m^3}$, ρ = 4400 $\mathrm{kg/m^3}$, N_T = 5 kN /m, ρ_a = 0.089 $\mathrm{kg/m^3}$, Ma = 5, a_∞ = 295 m /s, and l_x = l_y = 1 m. The acoustic load $p_{acoustic}$ is modeled by a random pressure. The initial displacement and velocity are set to zero. Supposing that the panel is simply supported, the boundary conditions are expressed as

$$\begin{cases} w(0,y,t) = w(l_x,y,t) = 0\,,\,(y,t) \in [0,l_y] \times [0,T] \\ w(x,0,t) = w(x,l_y,t) = 0\,,\,(x,t) \in [0,l_x] \times [0,T] \\ \dfrac{\partial^2 w(0,y,t)}{\partial x^2} = \dfrac{\partial^2 w(l_x,y,t)}{\partial x^2} = 0\,,\,(y,t) \in [0,l_y] \times [0,T] \\ \dfrac{\partial^2 w(x,0,t)}{\partial y^2} = \dfrac{\partial^2 w(x,l_y,t)}{\partial y^2} = 0\,,\,(x,t) \in [0,l_x] \times [0,T] \end{cases} \tag{41}$$

Table 4
Performance comparison of different optimizers.

Case No.	Optimizer	Relative L_2 error	Number of iterations	Maximum allocated memory (MB)	Training time (s)
4	Adam and L-BFGS	6.06E-05	3.41E+04	195.76	439
15	Adam	3.07E-03	3.41E+04	192.93	392
16	L-BFGS	4.88E-04	1.47E+04	195.76	195

Table 5
Performance comparison of different methods for free vibration analysis of a beam.

Case No.	Method	Hard constraints			Auxiliary functions	Relative L_2 error	Number of iterations
		Boundary displacement	Initial displacement	Initial velocity			
1	STD-PINN	×	×	×	N /A	7.48E-01	3.47E+04
2	AT-PINN	×	×	×	N /A	3.65E-03	1.12E+05
3	AT-PINN-HC1	√	×	×	ψ_P	1.28E-03	1.01E+05
4	AT-PINN-HC1	√	×	×	ψ_T	**2.21E-04**	**2.60E+04**
5	AT-PINN-HC1	√	×	×	ψ_E	4.40E-03	7.27E+04
6	AT-PINN-HC1	√	×	×	ψ_L	5.18E-03	6.18E+04
7	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1P}	7.69E-03	2.54E+04
8	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1T}	2.44E-03	2.72E+04
9	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1E}	**4.55E-04**	2.88E+04
10	AT-PINN-HC2	√	√	×	ψ_T, ϕ_{1L}	6.47E-04	2.55E+04
11	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2P}	2.05E-02	3.58E+04
12	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2T}	8.11E-03	3.11E+04
13	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2E}	**1.54E-03**	2.46E+04
14	AT-PINN-HC3	√	√	√	ψ_T, ϕ_{2L}	5.60E-03	2.62E+04

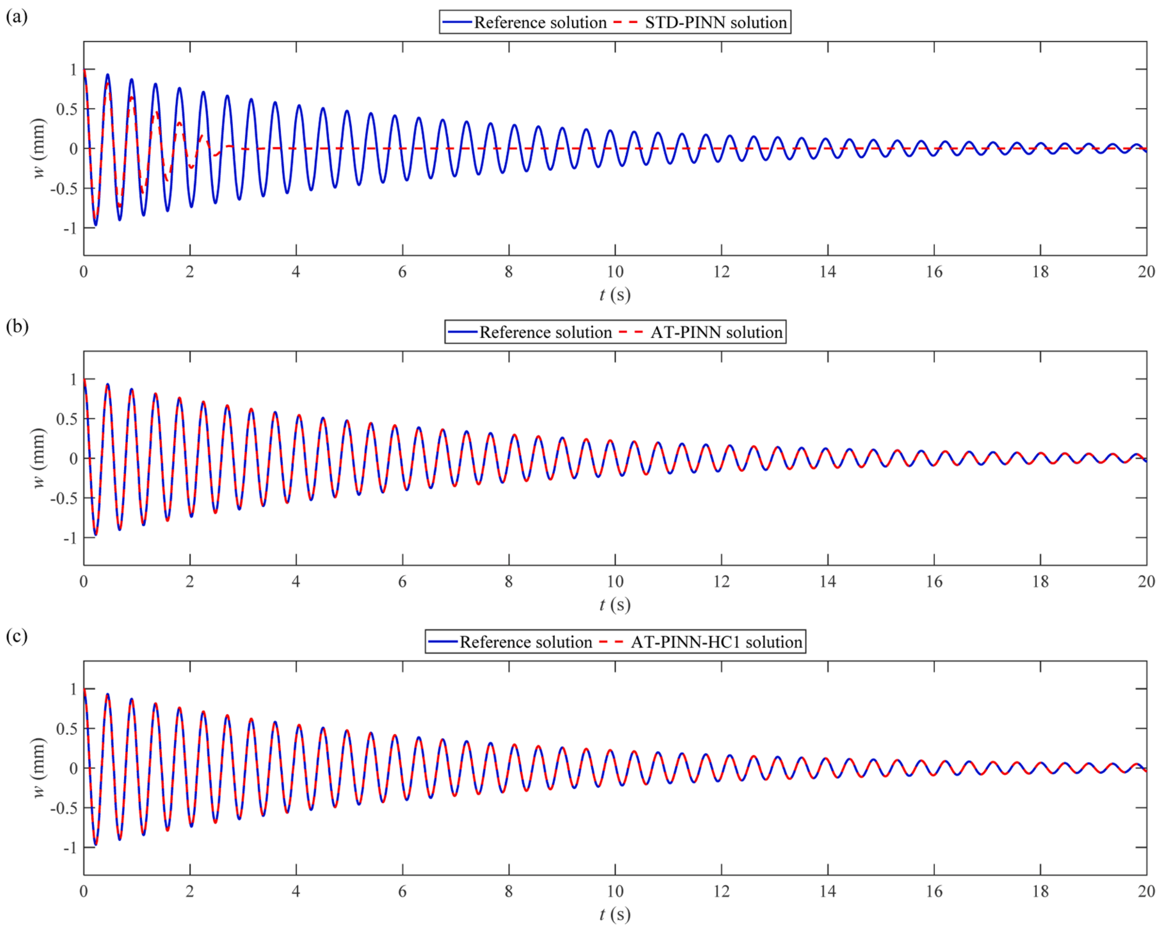

Fig. 13. Comparison of the reference solution and predicted solutions of (a) STD-PINN, (b) AT-PINN and (c) AT-PINN-HC1 with ψ_T at the midpoint of the free-vibration beam.

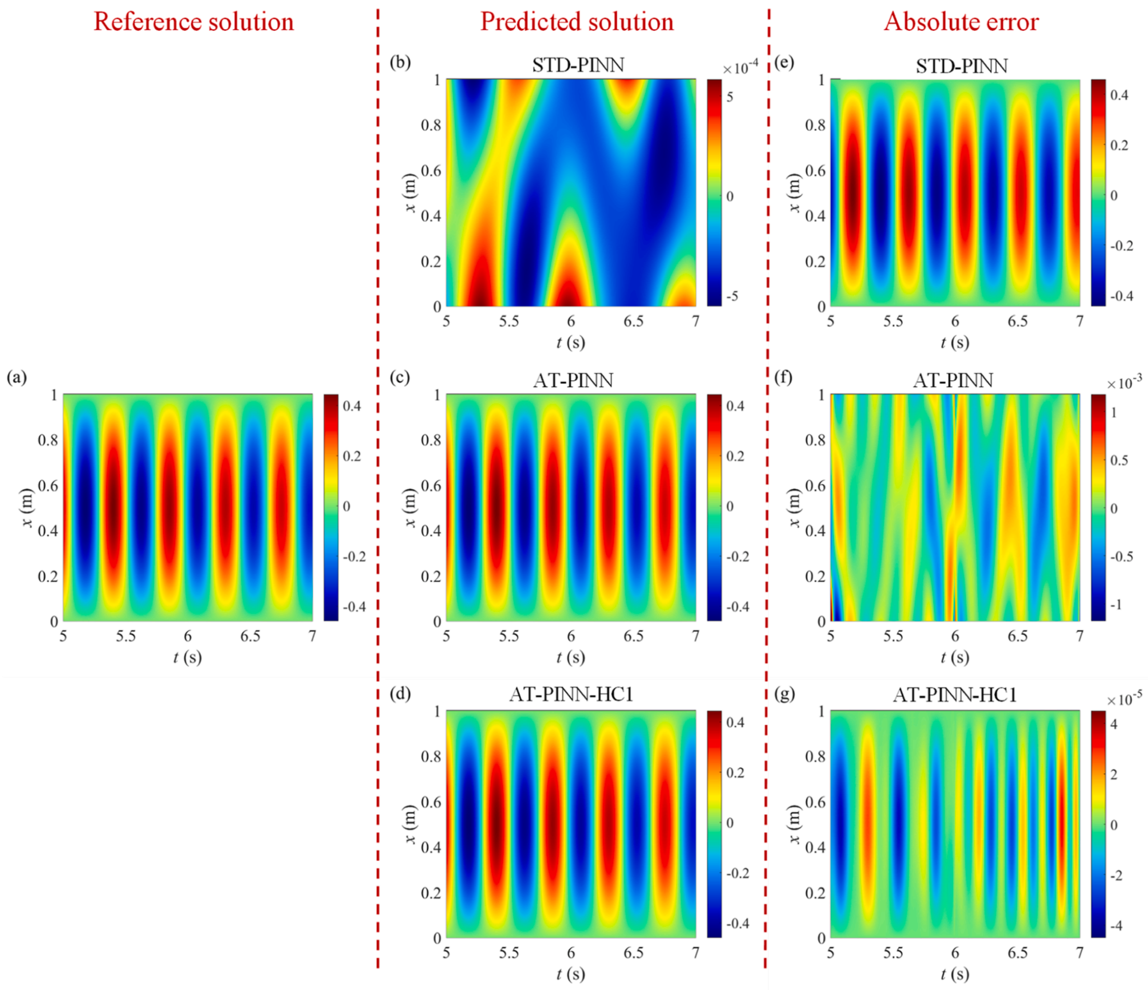

Fig. 14. Contours of solutions and errors of the free-vibration beam when 5 s $\leq t \leq$ 7 s: (a) reference solution; (b)-(d) predicted solutions of STD-PINN, AT-PINN and AT-PINN-HC1 with ψ_{T}; (e)-(g) absolute errors of STD-PINN, AT-PINN and AT-PINN-HC1 with ψ_{T}.

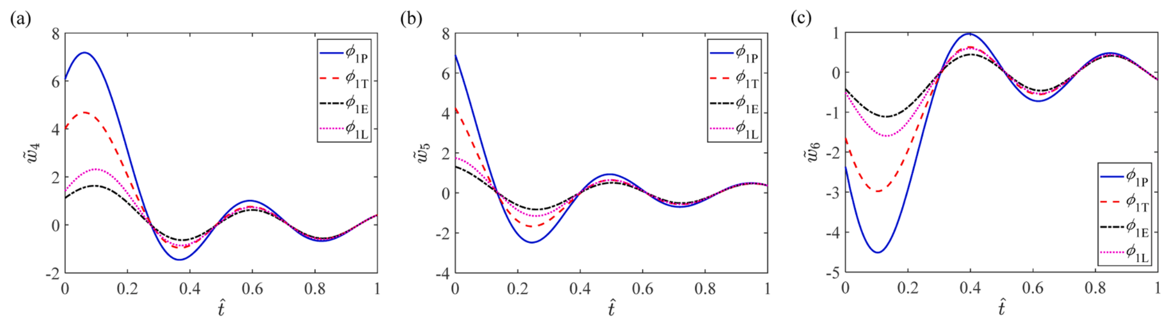

Fig. 15. Comparison of the output $\widetilde{w}_n$ before hard constraints with different types of ϕ_1 in the (a) 4th, (b) 5th and (c) 6th time segments at the midpoint of the free-vibration beam by using AT-PINN-HC2.

To implement hard constraints on the boundary displacement, the trigonometric auxiliary function ψ_{T} is constructed as

$$\psi_{\mathrm{T}} = \sin(\pi\hat{x})\sin(\pi\hat{y}) \tag{42}$$

The vibration behaviour of the panel with $T = 50$ s is analyzed by the AT-PINN and AT-PINN-HC methods. The entire time domain is divided into time segments with a time interval of $\Delta T = 0.2$ s in the implementation of these methods. In each time segment, the

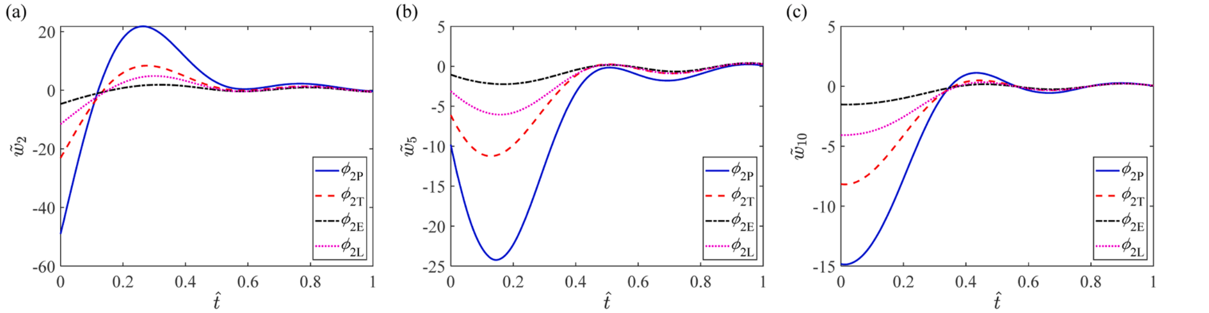

Fig. 16. Comparison of the output $\tilde{w}_n$ before hard constraints with different types of ϕ_2 in the (a) 2nd, (b) 5th and (c) 10th time segments at the midpoint of the free-vibration beam by using AT-PINN-HC3.

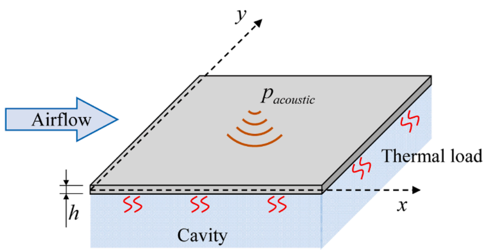

Fig. 17. Schematic diagram of a supersonic vehicle skin panel subjected to aerodynamic, thermal and acoustic loads.

numbers of sample points for the PDE, the boundary and initial conditions are set to $N_p = 16400$, $N_b = 8200$ and $N_i = 2500$, respectively. The weights of the loss components are given as $\lambda_p = 0.1$, $\lambda_{b1} = 5000$, $\lambda_{b2} = 100$, $\lambda_{i1} = 1000$ and $\lambda_{i2} = 10$. After the neural networks are trained, the solutions on 4×10^8 points uniformly sampled in the entire spatiotemporal domain are predicted and compared with the reference solutions to calculate the relative L_2 error. Fig. 18 illustrates the vibration response at the midpoint of the panel. Although vibration response of the panel fluctuates violently, the predicted solutions of the three AT-PINN-HC methods all show excellent coincidence with the reference solution. To verify the validity of the proposed hard-constraint strategies for such a complex vibrational problem further, the relative L_2 errors and numbers of iterations of the three AT-PINN-HC methods are compared with AT-PINN in Table 6, and some representative contours of solutions and errors of these methods are compared in Fig. 19. It can be observed that the AT-PINN-HC methods all perform better than AT-PINN in terms of solution accuracy and training efficiency. For instance, compared with AT-PINN, AT-PINN-HC1 can reduce the relative L_2 error by about 81 % and reduce the number of iterations by about 55 %.

3.4. *Wind-induced vibration analysis of a standing glass plate*

Glass windows are commonly used as external facades in building construction. Large glass panes in tall buildings are exposed to strong wind loads, resulting in intense vibrations that may pose safety hazards. In this example, we consider the vibration problem of a standing glass plate. According to the previous results, AT-PINN-HC1 with ψ_T has the best solution accuracy. In this section, this method is applied to the wind-induced vibration analysis of a standing glass plate as shown in Fig. 20. Taking the effect of gravity into consideration, the equation of motion of a vertical plate is expressed as [66,67]

$$\frac{Eh^3}{12(1-\nu^2)}\nabla^4 w + \frac{\partial}{\partial y}\left[(l_y - y)\rho h g \frac{\partial w}{\partial y}\right] + c\frac{\partial w}{\partial t} + \rho h \frac{\partial^2 w}{\partial t^2} = p_{wind}, \quad (x, y, t) \in (0, l_x) \times (0, l_y) \times (0, T] \tag{43}$$

where E is the elastic modulus, h is the thickness, ν is the Poisson's ratio, g is the gravitational acceleration, c is the viscous damping coefficient, ρ is the glass density, p_{wind} is the wind load, T is the simulation duration, l_x and l_y are the horizon and vertical lengths of the glass plate. In this example, these parameters are given as $E = 70$ GPa, $h = 5$ mm, $\nu = 0.3$, $c = 200\ \mathrm{N \cdot s/m^3}$, $g = 9.8\ \mathrm{m/s^2}$, $\rho = 2500\ \mathrm{kg/m^3}$, $T = 100$ s, and $l_x = l_y = 1$ m.

In this example, the wind load data were obtained from a wind tunnel test [68] and considered as a spatially uniform pressure. The time-history responses and the corresponding frequency spectrum of the wind data are presented in Fig. 21. The initial displacement and velocity are set to zero. Two boundary conditions are studied here. First, the glass plate is simply supported, and the boundary conditions with the corresponding auxiliary function ψ_T are the same as Eqs. (41) and (42). Second, the glass plate is clamped, so the displacement and slope on boundaries are all zero as follows:

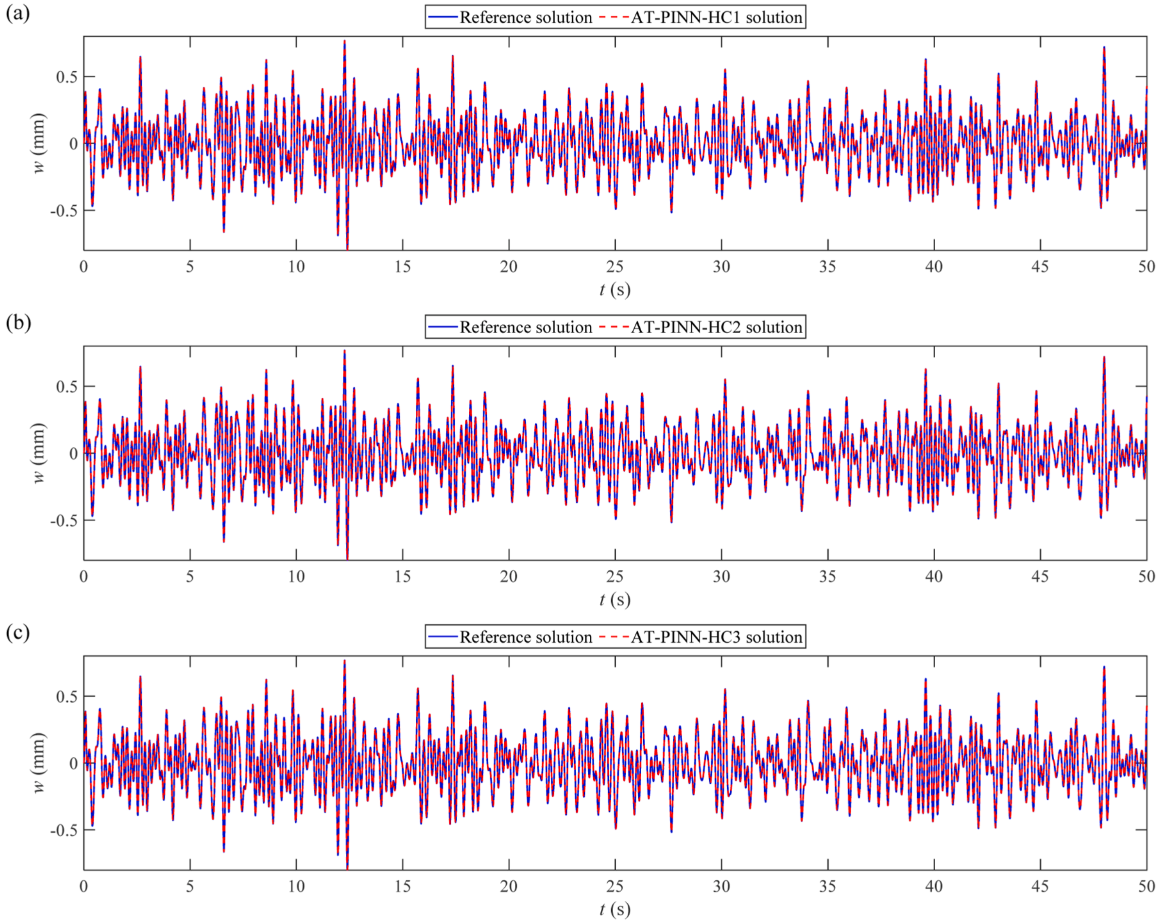

Fig. 18. Comparison of the reference solution and predicted solutions of (a) AT-PINN-HC1 with ψ_{T}, (b) AT-PINN-HC2 with ψ_{T} and $\phi_{1\mathrm{E}}$, and (c) AT-PINN-HC3 with ψ_{T} and $\phi_{2\mathrm{E}}$ at the midpoint of the supersonic vehicle skin panel.

Table 6
Performance comparison of different methods for vibration analysis of a supersonic vehicle skin panel.

Case No.	Method	Hard constraints			Auxiliary functions	Relative L_2 error	Number of iterations
		Boundary displacement	Initial displacement	Initial velocity			
1	AT-PINN	×	×	×	N /A	6.22E-03	4.60E+06
2	AT-PINN-HC1	√	×	×	ψ_{T}	1.21E-03	2.08E+06
3	AT-PINN-HC2	√	√	×	ψ_{T}, $\phi_{1\mathrm{E}}$	1.33E-03	2.01E+06
4	AT-PINN-HC3	√	√	√	ψ_{T}, $\phi_{2\mathrm{E}}$	1.63E-03	2.21E+06

$$\begin{cases} w(0,y,t) = w(l_x,y,t) = 0\,, (y,t) \in [0,l_y] \times [0,T] \\ w(x,0,t) = w(x,l_y,t) = 0\,, (x,t) \in [0,l_x] \times [0,T] \\ \dfrac{\partial w(0,y,t)}{\partial x} = \dfrac{\partial w(l_x,y,t)}{\partial x} = 0\,, (y,t) \in [0,l_y] \times [0,T] \\ \dfrac{\partial w(x,0,t)}{\partial y} = \dfrac{\partial w(x,l_y,t)}{\partial y} = 0\,, (x,t) \in [0,l_x] \times [0,T] \end{cases} \tag{44}$$

For this clamped boundary, we construct the trigonometric auxiliary function ψ_{T} as

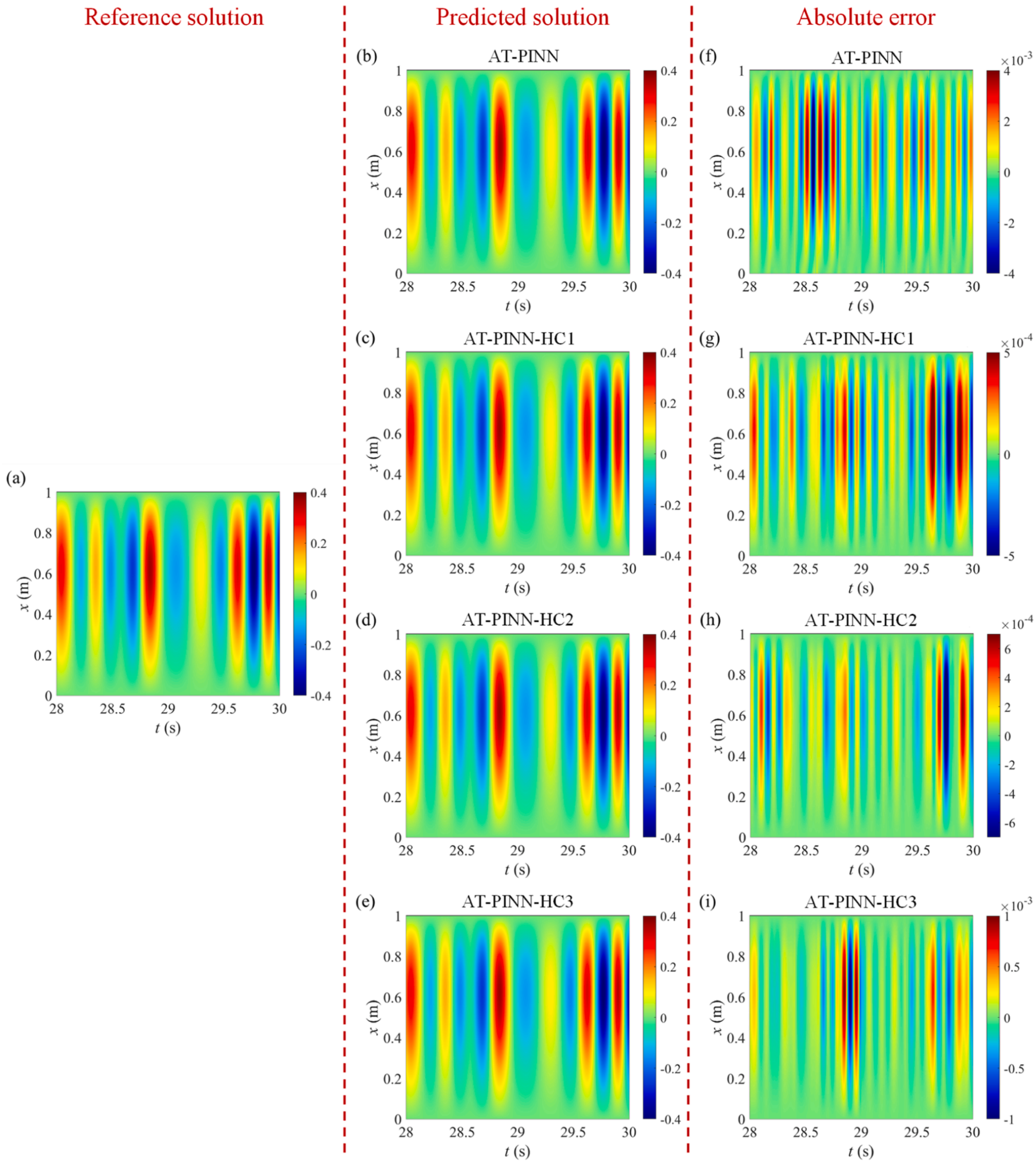

Fig. 19. Contours of solutions and errors of the supersonic vehicle skin panel at $y = 0.5$ m and $28\text{ s} \leq t \leq 30$ s: (a) reference solution; (b)-(e) predicted solutions of AT-PINN, AT-PINN-HC1, AT-PINN-HC2 and AT-PINN-HC3; (f)-(i) absolute errors of AT-PINN, AT-PINN-HC1, AT-PINN-HC2 and AT-PINN-HC3.

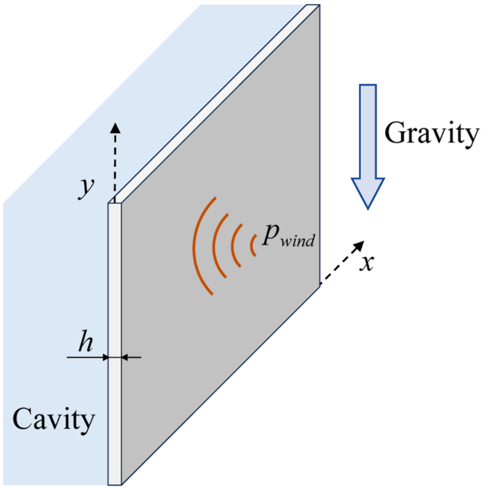

Fig. 20. Schematic diagram of a standing glass plate subjected to wind load.

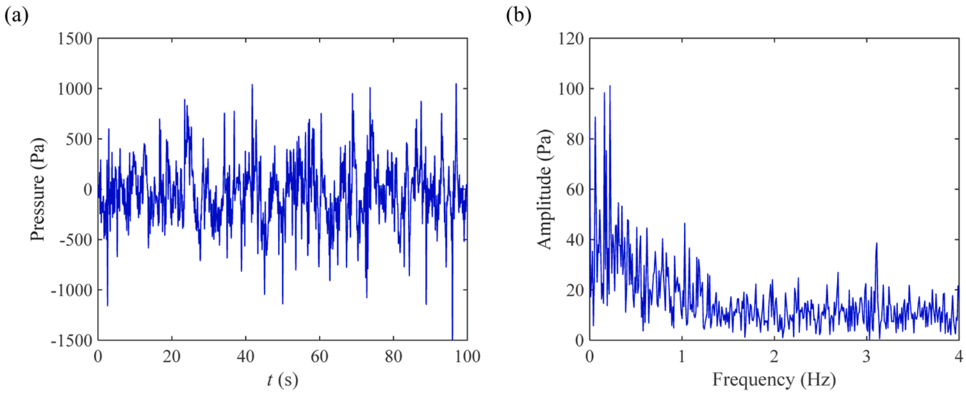

Fig. 21. (a) Time-history responses and (b) its corresponding frequency spectrum of the wind data [68].

$$\psi_{\mathrm{T}} = \sin^2(\pi\hat{x})\sin^2(\pi\hat{y}) \tag{45}$$

which can impose hard constraints on both the displacement and slope on boundaries simultaneously.

To solve the wind-induced vibration problem of the glass plate by AT-PINN-HC1, the entire time domain is divided into time segments with a time interval of $\Delta T = 0.5$ s. In each time segment, the numbers of sample points for the PDE, boundary and initial conditions are set to $N_p = 20500$, $N_b = 10200$ and $N_i = 2500$, respectively. For the simply supported glass plate, the weights of the loss components are given as $\lambda_p = 0.01$, $\lambda_{b2} = 100$, $\lambda_{i1} = 10000$ and $\lambda_{i2} = 10$. While for the clamped glass plate, there is no boundary loss when using AT-PINN-HC1 with ψ_{T}, and the weights of the loss components are chosen as $\lambda_p = 0.0001$, $\lambda_{i1} = 10000$ and $\lambda_{i2} = 10$. A reference solution is also obtained by the traditional Galerkin method and the Runge−Kutta method to verify the accuracy of the AT-PINN-HC1 solution. Fig. 22 compares the vibration responses at the midpoint of the glass plate. Fig. 23 compares the instantaneous vibration shapes at $y = 0.5$ m and $t = 60$ s. It can be observed that the AT-PINN-HC1 solutions agree very well with the reference solutions for both the simply supported and the clamped glass plates for $t = 0 - 100$ s. The simply-supported and clamped boundary conditions can be well satisfied by the AT-PINN-HC1 method. As presented in Table 7, the relative L_2 errors of AT-PINN-HC1 for both boundary conditions are lower than 5E-03. These results demonstrate the flexibility and applicability of the proposed hard-constraint strategies for vibrational problems with different types of boundary conditions.

4. Concluding remarks

In this paper, we have proposed and integrated three types of hard-constraint strategies for vibrational problems into a series of AT-PINN-HC methods. These strategies include: (i) hard constraints on boundary displacement; (ii) hard constraints on both boundary and initial displacement; and (iii) hard constraints on boundary displacement, initial displacement, and velocity. In addition to the commonly used polynomial and power functions for hard constraints, we introduced three different types of auxiliary functions based

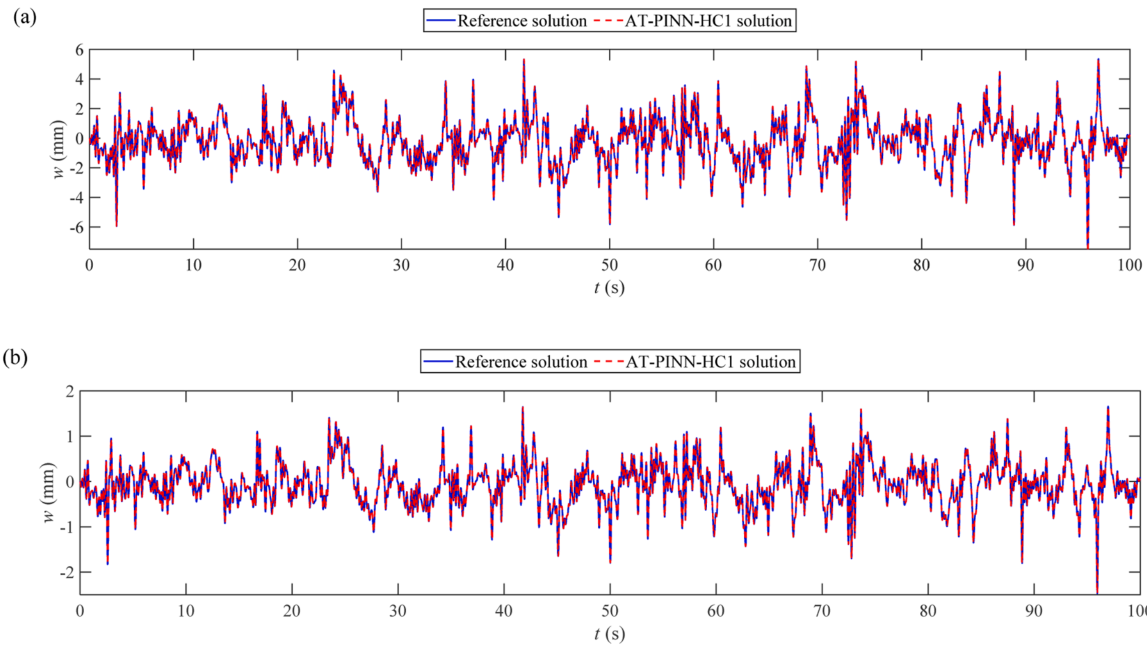

Fig. 22. Comparison of the reference solutions and predicted solutions of AT-PINN-HC1 with ψ_T at the midpoint of the standing glass plate with (a) simply-supported boundary conditions; and (b) clamped boundary conditions.

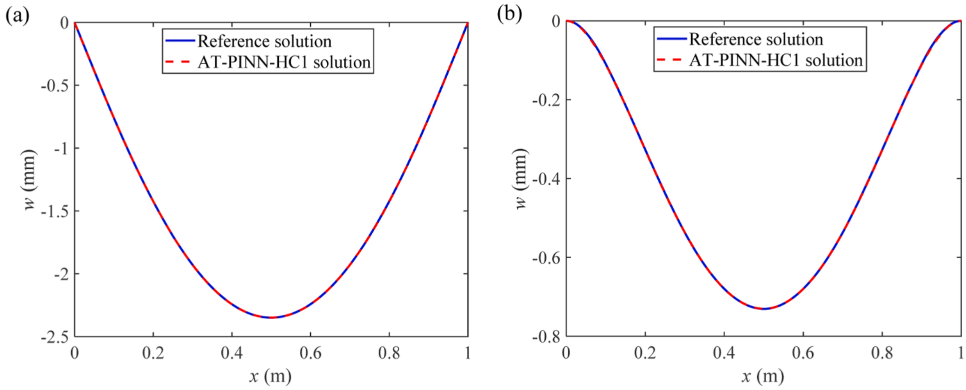

Fig. 23. Comparison of the reference solutions and predicted solutions of AT-PINN-HC1 with ψ_T at $y = 0.5$ m and $t = 60$ s for the standing glass plate with (a) simply-supported boundary conditions; and (b) clamped boundary conditions.

Table 7
Performance of AT-PINN-HC1 with ψ_T for wind-induced vibration analysis of a standing glass plate with different boundary conditions.

Case No.	Boundary condition	Method	Auxiliary functions	Relative L_2 error	Number of iterations
1	Simply supported	AT-PINN-HC1	$\psi_T = \sin(\pi\hat{x})\sin(\pi\hat{y})$	2.97E-03	3.81E+06
2	Clamped	AT-PINN-HC1	$\psi_T = \sin^2(\pi\hat{x})\sin^2(\pi\hat{y})$	4.53E-03	2.89E+06

on the trigonometric, exponential, and logarithmic functions for each strategy. To validate the effectiveness, applicability, and advantages of the proposed AT-PINN-HC methods, we conducted vibration analyses on an Euler−Bernoulli beam, a supersonic vehicle skin panel under multi-physics loads, and a standing glass plate under wind load. The main conclusions are summarized as follows:

- The proposed hard-constraint strategies and auxiliary functions ensure that the boundary and initial conditions are strictly satisfied, thereby greatly improving both solution accuracy and training efficiency. The AT-PINN-HC methods accurately solve vibration problems in long-duration simulations, unlike the STD-PINN method. Compared to AT-PINN without hard constraints, AT-PINN-HC reduces the relative L_2 error by an order of magnitude and decreases the number of iterations by up to 78 %.

- Different boundary and initial conditions require different auxiliary functions for hard constraints. Polynomial and power functions, commonly used for hard constraints, are not optimal for vibrational problems. For boundary displacement constraints, the proposed trigonometric auxiliary function achieves the highest solution accuracy and training efficiency. For initial displacement and velocity constraints, the solution accuracy is related to the first and second derivatives of the auxiliary function. The proposed exponential auxiliary functions exhibit the highest first and second derivatives, resulting in the smoothest output variation. This characteristic facilitates a more straightforward learning of the mapping relationship between inputs and outputs compared to other auxiliary functions. Consequently, these exponential auxiliary functions are ideal for enforcing hard constraints on the initial displacement and velocity in terms of solution accuracy.
- Enforcing hard constraints on additional initial conditions does not automatically result in more accurate solutions. Among the three hard-constraint strategies, applying constraints exclusively to boundary displacement consistently produces the most accurate results, outperforming the approach that applies constraints to both boundary and initial displacement. The least accuracy is noted when constraints are imposed on boundary displacement, initial displacement, and velocity simultaneously. This is due to the fact that adding more initial condition constraints complicates the mapping relationship between inputs and outputs, making it harder to train the neural network effectively, which in turn decreases solution accuracy.
- Applying hard constraints can simplify the loss function components, facilitating the weight adjustment of them. In practical applications, it is crucial to weigh the advantages and disadvantages of hard constraints and select the appropriate strategy and auxiliary functions accordingly.

Going beyond vibrational problems, the present work is expected to be applicable to other time-dependent PDEs and inspire researchers to develop novel hard-constraint strategies and auxiliary functions. Despite these advancements, the proposed AT-PINN-HC methods have certain limitations. One such limitation is that the transfer learning technique used in AT-PINN-HC is effective only when the solutions in neighboring time segments are similar. In cases of complex vibrational problems, where the vibration responses vary considerably between adjacent time segments, this transfer learning approach does not enhance the training process. Furthermore, these methods are confined to structures with regular geometric shapes, which motivates us to pursue the extension of AT-PINN-HC to incorporate complex geometries in the future.

CRediT authorship contribution statement

Zhaolin Chen: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Siu-Kai Lai:** Writing – review & editing, Methodology, Investigation, Conceptualization, Supervision. **Zhicheng Yang:** Writing – review & editing, Software, Methodology, Investigation. **Yi-Qing Ni:** Writing – review & editing, Methodology, Investigation. **Zhichun Yang:** Writing – review & editing, Validation, Investigation. **Ka Chun Cheung:** Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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